Loan growth and its impact on non-performing loans and profitability
Understanding the trade-off - A study of the Indian banking sector

Thesis submitted in partial fulfillmentfor

the Award of Degree of

Doctor of Philosophy

by

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PM-19-12 **2024**

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Dedicated To

Lord Abhaydata,

Maa Vaishno devi

My mother and my husband

For their blessings, never-ending support, and belief in my efforts.

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SHWETA GUPTA

ABSTRACT

The last one and a half decades have been tough for the global economy which witnessed the global financial crisis (GFC) during 2008-2009. The aftereffects of the GFC could be seen in the balance sheets of financial institutions globally. Indian Banks are not different either. The Indian Banks suffered the delayed effects of GFC in 2014-15 when the apex bank of India, The Reserve Bank of India (RBI), decided to do a random Asset Quality Review (AQR) of the banks, suspecting that many banks were not strictly following the Income Recognition and Asset Classification (IRAC) norms. The AQR revealed a tonne of bad loans hidden in the banks' balance sheets. The Non-Performing Asset(NPA) ratio rose from 4.5 % to a staggering 7.5 % in 2016,9.3 % in 2017, and 11.2 % in 2018 in three years. The increased burden of bad loans brought a slowdown in loan growth.

The impact of increased Non-Performing Loans (NPLs) on lending activities can be seen in the loan rate since 2016. The slowdown in lending activities continued for six to seven years; now, after a half-decade of balance sheet clean-up prompted by the RBI's AQR, Indian banks are back on track for increased loan growth. The loan growth has increased from 5.4 % and 6% in 2020-21 and 2019-20, respectively, to almost double in 2021-22 to 11.4 %. Bank loans are a very important and integral part of the income of the banks, and it is also a necessary impetus for economic activities; hence they cannot be put on hold for a long period. But it is also important to understand that bank loans always carry a certain amount of default risk. This default risk depends on economic and market conditions and the due diligence with which a loan is sanctioned. This default risk brings into the picture the trade-offs between loan growth, NPLs, and bank profitability. Our study highlights the importance of understanding how loan growth

can manifest increased interest income, which increases profitability but also carries a risk of increased NPLs, which proves detrimental to the bank's profits. The world economy is once again going towards a slowdown. The events starting in the United States (U.S.) banking industry indicate the beginning of a crisis that might have a global impact. Thus, the banks must understand the behaviour of these important variables and formulate a credit policy accordingly to minimize the risk.

We have taken a sample of 12 public sector banks and 17 private sector banks for our study. Financial ratios of these banks for the period 2006-2021 are used to perform the analysis. Growth in advances is used as an indicator of loan growth. Return on assets (ROA) and Ratio of Net Non-Performing Assets to Gross Advance (NNPAtoGA) are used as indicators for profitability and NPLs, respectively. Also, some bank-specific and macroeconomic variables are added to our model as control variables. To overcome the endogeneity issues in the variables we have used the System Generalised Method of Moment(SYS-GMM) estimation as a first step to establish and understand the nature of the relationship between the variables. Next, we use Structured equation modelling to establish a mediating role of the NPLs between the relationship of loan growth and profitability.

The study's findings show a significant relationship between loan growth, NPLs, and profitability. On one side, loan growth increases profitability but also NPLs, eventually decreasing profitability. We use quadratic terms of loan growth to test the relationship between loan growth and Profitability, loan growth, and NPLs.We find that loan growth has a U-shaped relationship with the NPLs and an inverse U-shaped relationship with the profit. The study result indicates moderate loan growth as a key to steady and stable growth of the banking industry in

India. At the same time, studying the simultaneous relationship between our variables of interest. We find that NPLs very significantly mediate the relationship between the two variables.

The period of boom and recession is a part of the economic cycle. These different phases in the economic cycle affect the financials of all the businesses in the economy, which also affects the banking industry in a country. The major portion of this impact comes from the loans part of the bank. Excessive lending during the boom phase due to increased economic activities converts to bad loans in times of recession when it becomes difficult for the businesses to survive, let alone repay the bank loans. This study will help the bank's management to understand that although loan growth is beneficial for the bank, after a certain point the negatives start outweighing the positive impacts of the lending. The increased NPLs due to excessive lending put pressure on the bank's profits in terms of increased provisioning. This study will focus towards understanding the trade off between the negative and positive manifestation of loan growth .It will help the banks to formulate credit policy keeping in mind the impact that it can have on NPLs and profitability and thus outweighing the negative impact by the positive.

The issue of the simultaneous impact of loan growth on NPLs and profitability has not been studied in the Indian Banking sector. Also, this study adds to the present literature by studying the mediation effect of the NPLs on the loan growth and profitability relationship.

Keywords – Loan Growth, Non-Performing Loans, Profitability, Non – Linear relationships, Generalised Method of Moments, Structural Equation Modelling

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ABBREVIATIONS

AQR	Asset Quality Review
CDR	Corporate Debt Restructuring
CPI	Consumer price index
DV	Dependent Variable
GCC	Gulf Cooperation Council
GDP	Gross Domestic Product
GFC	Global Financial Crisis
GMM	Generalised Method of Moment
GNPA	Gross Non-performing Assets
IMF	International Monterey Fund
INF	Inflation
IRAC	Income Recognition and Asset Classification
IV	Independent Variable
LLP	Loan loss provision
LNTA	Log of total assets
MENA	Middle East and North Africa
MSEs	Micro and Small Enterprises
NII	Non-Interest income
NIM	Net Interest Margin
NNPAtoGA	Ratio of Net Non-Performing Assets to Gross Advance
NPA	Non-Performing Asset
NPL	Non-Performing Loan
OD/CC	Overdraft or Cash Credit
OLS	Ordinary Least squares
PSL	Priority sector lending
RBI	Reserve Bank of India
RID	Ratio of the indirect effect to the direct effect

RIT	Ratio of the indirect effect to the total effect
ROA	Return on assets
ROE	Return on Equity
SYS-GMM	System Generalised Method of Moment
TA	Total Asset
TL	Total loans
U.S.	United States
WoS	Web of Science
SEM	Structural Equation Modelling
-ve	Negative
+ve	Positive
PSB	Public Sector Banks
Pvt Banks	Private Sector Banks

Chapter 1

Introduction

The Indian banking sector has evolved through many changes and phases since its inception. Post-independence in 1969, the nationalization of the banks was done to sync the banking industry goals with the social goals of the economy. The present structure of the Indian Banks originated from the New Economic Policy of 1991, which promoted globalization, privatization, and liberalization (Sanyal & Shankar, 2011). Post-1991, the banks are now open to business globally; however, they are still obligated to assist the government in achieving some social objectives.

1.1 Importance of bank lending in the nation's economic growth and bank profits.

Banks play a pivotal role in developing the economy(Menicucci & Paolucci, 2016). Over the years, various ups and downs have occurred in the banking industry globally. One of the major reasons for the banking woes has always been rising delinquencies in the loans (Gavin & Hausmann, 1996). A major portion of the country's financial assets is held by banks, which play the role of the primary lenders in the country's financial system. Government-controlled banks have played a crucial role in India's development plan since the 1970s by providing financing to areas that are emphasized by governments, such as infrastructure and agriculture, as these sectors require financial support from the banking sector in the form of formal credit(Gaur & Mohapatra, 2021b). (Sutton, 2021) As NPL ratios started to rise sharply in the middle of the 2010s, RBI implemented several steps to address balance sheet problems in Indian banks. The RBI restricted lending by select banks to ease pressure from the increasing load of NPAs. (Viral V. Acharya, 2018) The RBI also introduced a restructuring and NPA resolution framework to help Banks lower their levels of NPAs. (RBI, 2019).

1.2 Non-performing loans -A threat to the bank's growth.

An increasing level of NPLs brings the economy to a phase of slow credit growth. However, Bank loans are an important impetus for economic growth. It is also a core function of commercial banks. Banks mobilize funds from depositors and channel those funds toward borrowers in need. Bank lending activities support economic growth, drive investment, and enable individuals and businesses to achieve financial goals. Thus, it is safe to conclude that bank lending is a very important activity of the banks that can never be put on hold.

Lending activity is also complex as it requires expertise at both the pre-sanction and post-sanction levels. Increased rate of lending affects the appraisal quality, and also increased pace of lending by the banks impacts the economy, as over-lending can lead to an overextended credit supply, which exceeds the borrower's repaying capacity, leading to default (Dang, 2019a). Default in an instalment repayment is the first step towards an asset becoming a non-performing asset (NPA). NPAs directly affect profitability, as banks must make provisions at an increased rate on such assets per RBI directives. The indirect impact of NPAs on profitability is also studied (Adeolu, 2014). It has been mentioned that the increasing NPAs lower the bank ratings, and additional costs are incurred to recover such NPAs, reducing the profit.

1.3 Global economic scenario and its impact on the Indian banking sector

The Global financial crisis (GFC) of 2008, which started in the USA but became a global phenomenon, can be marked as an event that brought the significance of the banking system's health to the fore. The GFC, 2008 is an example of prioritizing profit motive at the cost of due diligence and steady growth. The event made the world realize how risk-based lending by banks without proper safeguards can lead to a massive breakdown in the banking sector and

the economy. The Indian banking sector withstood the shock of the global slowdown because of the Reserve Bank of India (RBI) intervention. However, the effects of the same could be seen in 2015 when RBI suspected that many of the banks were not adhering to the IRAC (Income recognition and asset classification) norms. RBI resorted to asset quality review to check the window dressing done by the banks behind the veil of corporate debt restructuring packages. This signifies that the crisis in the Indian banking industry is somehow related to the GFC of 2008. The following three figures show the financial stages that the Indian banking sector has been going through in the past one and a half decades. We can see in Figure 1.1 and Figure 1.2 that the Indian banking sector did not see an immediate impact on the GFC 2008 as there was a constant improvement in the Return on Assets (ROA) and Gross Non-performing Assets (GNPA) ratio. However, we see a delayed impact from 2014-15 onwards, where all the ratios see a downturn, which can be attributed to the asset quality review by the RBI in 2014-15 to ensure proper classification of standard and sub-standard assets. The drive was meant to clean the bank's balance sheets, which, according to the experts, were window-dressed behind the veil of RBI's CDR policy at that time.

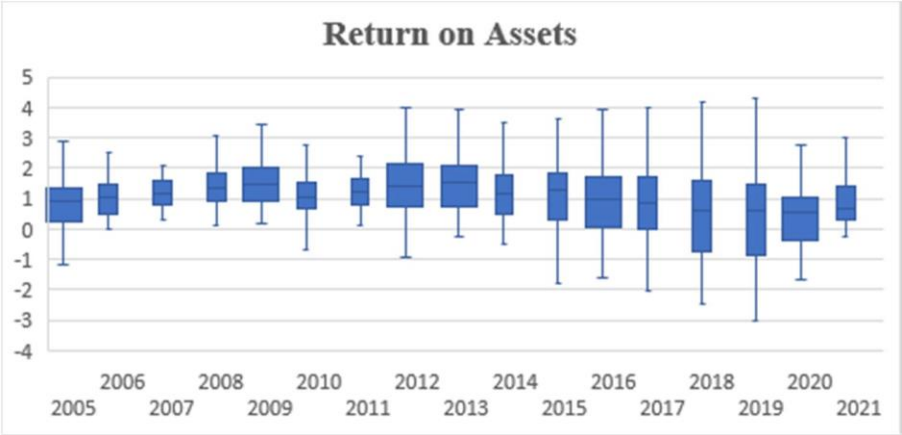

Figure 1.1: Return on Assets Indian Banking Sector 2005-2021

Source – Author's compilation from RBI Data

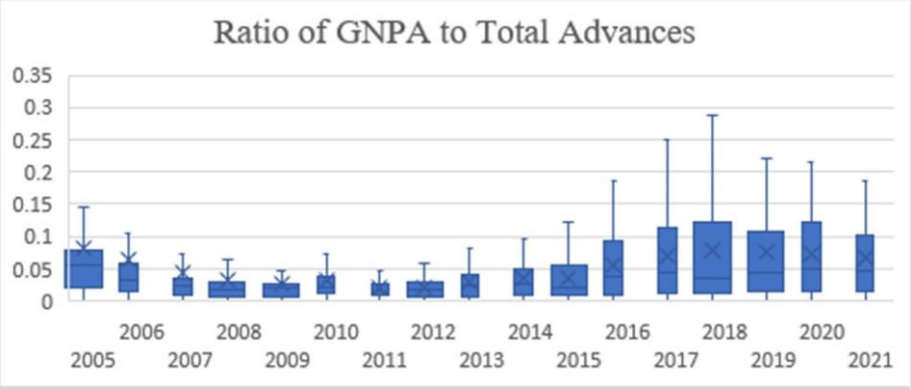

Figure 1.2 GNPA Indian Banking Sector 2005-2021

Source – Author's compilation from RBI data

1.4 Motivation for the Study

"History does not repeat itself, but it rhymes" – Mark Twain

This quote very beautifully explains that events happen repeatedly. They may not be of the same nature, but they occur in a similar pattern. It is thus very important to understand this pattern and plan accordingly to avoid a crisis.

As per the RBI's Report on Trend and Progress of Banking in India 2021-22, the loan growth has increased from 5.4 % and 6% in 2020-21 and 2019-20, respectively, to almost double in 2021-22 to 11.4 %. The increased lending is good news as finance is the impetus for all economic activities.

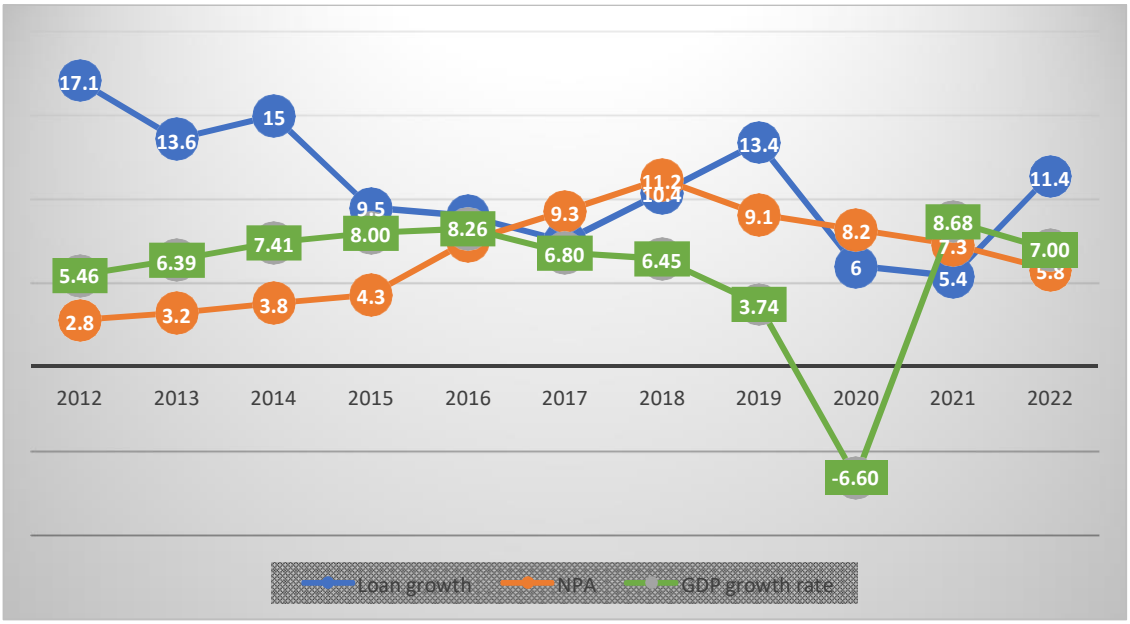

Figure 1.3 Loan growth and Movement of NPLs in Scheduled commercial banks of India with the GDP growth rate

Source – Author's compilation from RBI Data

Figure 1.3 shows the loan growth, GDP growth, and movement of NPA in Indian banks over the last decade. The pattern of lending and the NPLs in the banking industry worldwide can be mapped with certain events in the nation's economic cycle. Bank lending increases in times of boom in the economy to meet the demand for credit, followed by an increase in the NPLS during the economic downturn. The reason for the same has been attributed to both – the lowering of the credit screening standards during the boom and the decreased disposable income to repay loans due to the recession (Caporale et al., 2014a). Economists predict a slowdown in economic growth due to a global economic slowdown in 2023 because of the rising political unrest, which again puts the banks at risk of increasing NPLs. The economic cycle follows a pattern with the period of expansion, peak, recession, depression, trough, and recovery following each other. The duration of each phase may vary depending on various other conditions. Hence, to safeguard the bank's business from the

economic cycle, banks must always act practically and maintain a healthy balance between growth and risk.

This dilemma of safeguarding the bank's health with the balance between growth and stability brings us to our research objective, which studies whether, in Indian Banking, loan growth is detrimental to the profitability of the bank or not and what amount of loan growth is beneficial for the bank's health.

1.5 Research objectives

It is imperative to strike a proper balance between growth, profitability, and the risk associated with loan growth, i.e., increasing NPLs. This paper will try to understand the trade-off between these three aspects- loan growth, NPLs, and profitability. We study the impact of loan growth on NPL and profitability while controlling for macroeconomic and bank-specific factors such as GDP growth, inflation, and bank size. Hence, the research objective of this study is to understand:

Research Questions:

RQ1 - How does loan growth impact the NPLs?

RQ2- How does loan growth impact profitability?

RQ3- How NPLs impact profitability?

RQ4- Do NPLs mediate the relationship between loan growth and profitability?

The remaining paper is designed in the following format: some existing literature on this topic will be reviewed in part 2, data and methodology will be discussed in part 3, and part 4 will be the result and discussion. The conclusion and implications of the study will be discussed in part 5. Finally, the future scope and limitations of the study will be discussed in part 6.

CHAPTER 2

LITERATURE REVIEW

Chapter 2

Literature Review

Banks accept deposits from the public for lending (Selgin, 2023). The interest earned from the money advanced forms a major share of the bank's income comes from. One way to increase the banks' profitability is to increase the quantum of loans, presuming that increasing the quantum of loans increases the profitability(Le, 2020a). Apart from being an income-earning vehicle, bank loans also carry l default risk (Kwan & Eisenbeis, 1997a). This makes lending a risky process. The ripple effect created by the Global Financial Crisis (GFC) 2008 is evidence that bank lending has a very deep impact on the macroeconomic environment, and also macroeconomic environment impacts both bank lending and profitability. As the bank management cannot control the macroeconomic environment, it is very important to understand the internal factors that impact the profitability while making lending decisions.

A credit boom period preceded the GFC (Ivashina & Scharfstein, 2010). However, as seen during the crisis, overlending by the banks can lead to an overextended economic credit suppl which exceeds the borrower's repaying capacity, leading to default (Dang, 2019a). NPLs directly affect profitability, as banks must make provisions at an increased rate on such assets (Balasubramaniam, 2011). Profitability is also indirectly impacted by NPLs, as increasing NPLs reduces the bank’s credit ratings and results in additional costs to recover such NPLs, which lowers profit. Over-lending is not the only reason for bad debts. Noncompliance with loan policy, inadequate project monitoring, incomplete knowledge of the borrower’s business activities, etc., also lead to bad debts(Abata, 2014). Despite the risks that loan growth carries, it forms an integral part of the bank business both as a major source of income for banks worldwide and as a source of financial impetus for economic activities.

The banking system is a very important and integral source of finance for a nation's economic activities. The banking crisis might drag the entire economy into a downturn. This study aims to understand the interrelationship between three fundamental aspects of any business, i.e., growth, profitability, and risk. We attempt to study these variables and how they interact in the banking sector, where growth in lending leads to increased profitability in the form of interest earned, which also carries a risk of default. Risk and Return have a unique relationship; if you want to earn better returns, you must bear a little more risk, but it is essential to have an optimum mix of risk and return (Eling et al., 2017). Risk creates a dual effect on profitability: beyond a threshold, the positive impact diminishes, and a further increase in risk reduces profitability. Therefore, firms prioritizing profitability with safety are more likely to achieve better growth than those prioritizing only growth. Thus triad of three variables must be researched to determine the type and degree of the effects that each one has on the others.

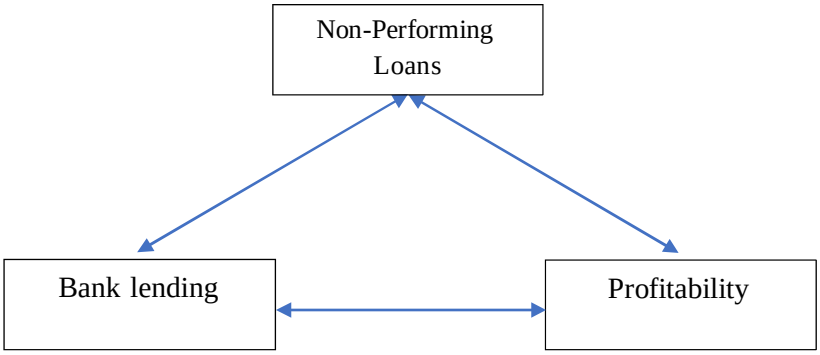

Figure 2.1 Triad of bank lending, bank profitability, and NPLs

We can find ample literature that has studied the relationship between loan growth and profitability, loan growth and NPLs, and NPLs and profitability and more such combinations, but we have not come across any research that combines the three factors and tried to

understand the interplay of relationships within these three aspects. As our study is intended to establish a relationship between bank lending, NPLs, and bank profitability to be a comprehensive study, we used combinations of possible synonyms of these three variables and searched the two databases Scopus and WoS, which gave us only a set of 27 articles that included all these three variables together, which seems to be too small a number to be able to understand the relationship between these variables together. Hence, we decided to review papers that study different combinations of any two variables from this triad and create a framework that explains the interrelationship of these variables, and we use the synonyms and all possible combinations of these variables for our search criteria. *Figure 2.2* shows the data retrieval process and the keywords used for the search.

Keywords are selected based on the review of some existing literature in this field of study. The keywords selected for Bank lending are bank lending (Bredl, n.d.; Ivashina & Scharfstein, 2010), loan growth (Dang, 2019; Foos et al., 2010b), and credit growth (Laidroo & Männasoo, 2013) Keywords selected for Non-performing loans are Credit risk (Agyei, 2017; Jreisat, 2020) Asset quality (Alqahtani et al., 2022) Non-performing loans (Ari et al., 2021) Keywords selected for Bank profitability are bank performance (Dang, 2019) and bank profitability (Mostak Ahamed, 2017a). The initial search was done using the PRISMA framework per the search criteria shown in Fig.2.2 on 19.12.22. The filter for the subject area, type, and language was applied. Subject Area for Scopus search was limited to “Economics, econometrics, and finance” business management and accounting” and “social sciences” and the Web of Science was limited to “business finance”, “Finance” and “management”. Document type is limited to “Article” and language is limited to “English”.

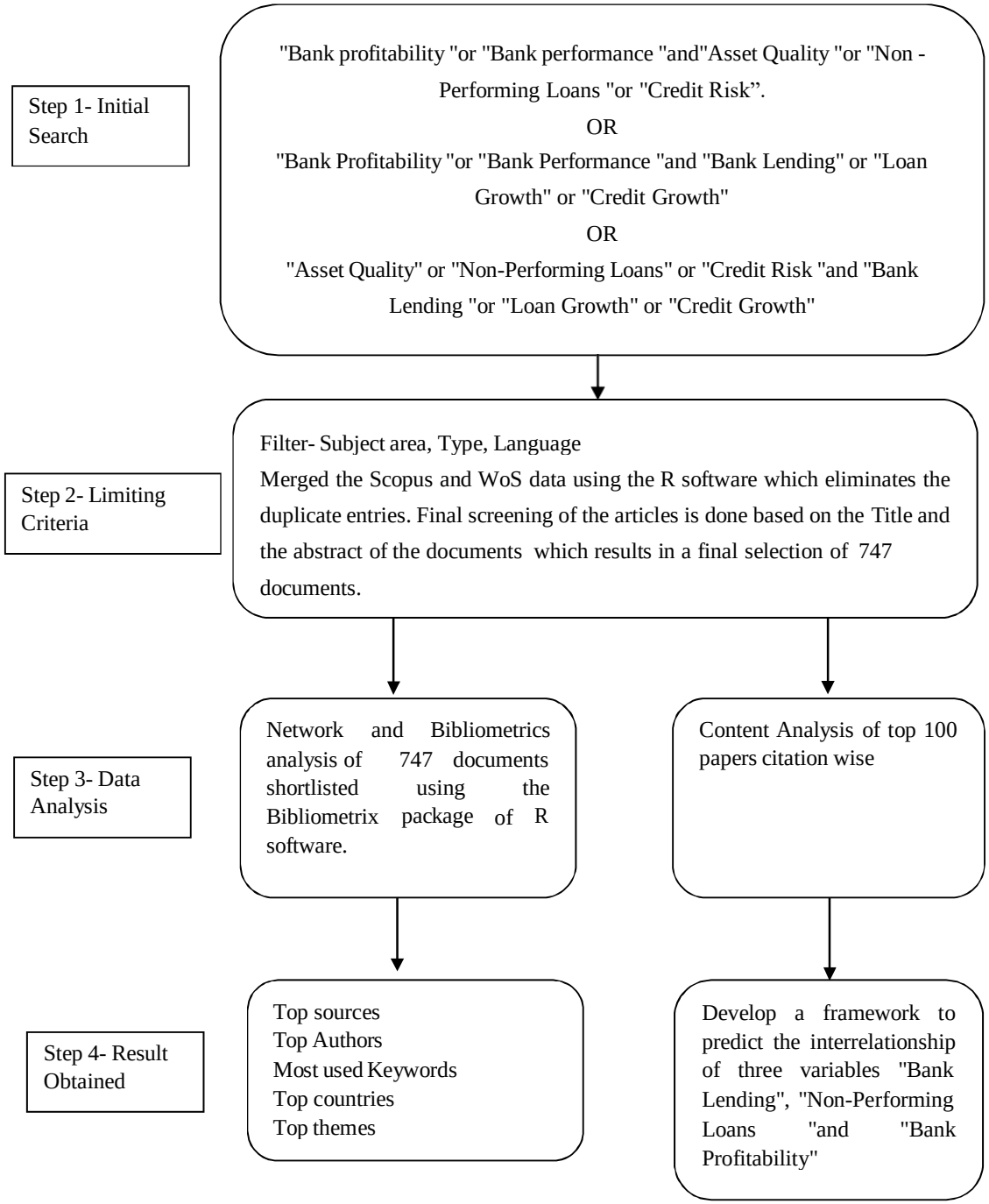

Figure 2.2 Data retreival process for SLR

This search criterion resulted in 559 articles from Scopus and 586 from WoS. The lists were merged using the *R* software, and the duplicate entries were eliminated using the same methods. The R software deleted the eighteen duplicate entries. Further, the remaining 1127 articles were screened by both the authors based on the title and the abstract. We finally arrived at a list of 747 articles on the topic. To better understand our field of study, we do a bibliometric analysis of these 747 articles, followed by the content analysis of some of the prominent papers in our area of research and a meta-analysis of the findings of the existing studies to synthesize a framework that helps us achieve our research objectives.

2.1 Bibliometric Analysis

The study used the Bibliometrix package of *R* software "Biblioshiny" and VOS Viewer for conducting the bibliometric analysis to understand the trend of publications and research in this area. We also conducted a content analysis of the top 100 papers based on the number of citations. The content analysis will provide future researchers with knowledge of the themes of the papers, the methodology used, and the type of data used.

The study by Donthu et al. (2021) gives an overview of how and when to conduct a bibliometric analysis. They have divided it into three parts: performance analysis, science mapping, and network analysis. In this study, we include parameters relevant to our study from all three parts of bibliometric analysis to understand the publication trends in this research area.

2.1.1 Performance Analysis

This section presents the results obtained from the bibliometric analysis performed on the data extracted from the Scopus and WoS databases. Firstly, we discuss the results of the performance analysis parameters.

2.1.1.1 Year-wise number of publications and publication growth analysis

Out of 747 papers, 695 were published after 2009, accounting for 88.22% of total publications. The first publication dates back to 1968. However, till 1999, the publication rate was very low, with only 17 papers published in 31 years. The number of publications started to increase in 2008, coinciding with the GFC, which highlighted the importance of the well-being of the banking sector and its importance for the whole economy. The effects of the subprime lending crisis in the United States were seen worldwide. This might have led to increased research on bank lending, NPLs, and profitability. The highest number of papers were published in 2020. The number of publications saw a minor fall in 2021, which again rose in 2022. Overall, the theme of bank lending, bank profitability, and NPLs seems to have become a popular area of research since the GFC of 2008, as it made people realize the importance of the banking system and its impact on the economy as a whole. Figure 2.3 shows a graph of several publications on this theme.

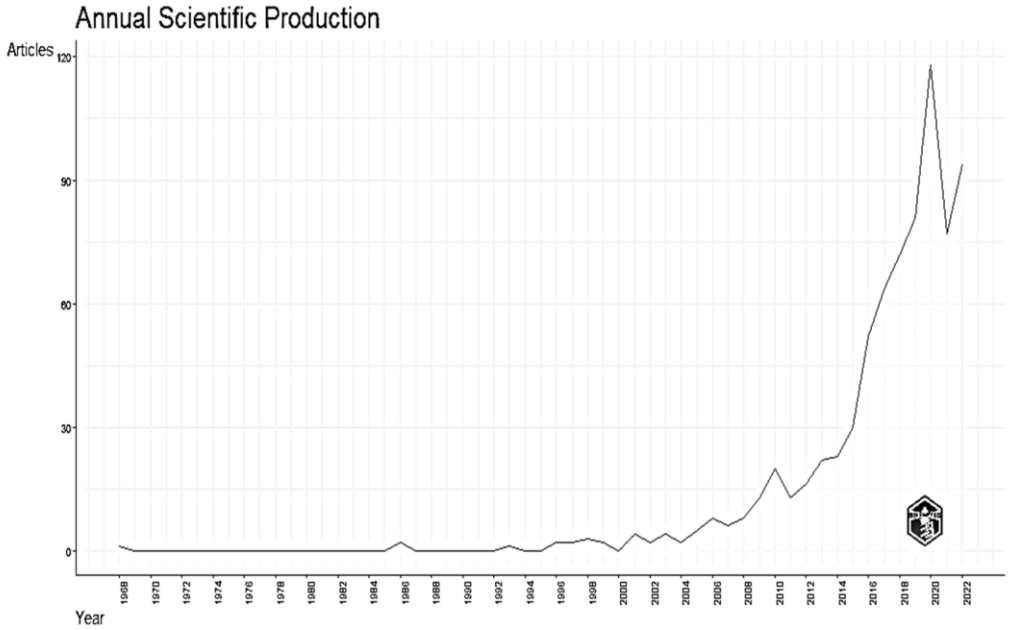

Figure 2.3 Year-wise number of publications

2.1.2. Science Mapping

2.1.2.1 Analysis of Thematic Evolution

An important part of science mapping in bibliometric analysis is an analysis of the changing themes of publication over time. The database extracted from Scopus and WoS dates back to 1968. We divide the total period into two-time slices with the cut-off point 2009. So, we have two time periods: 1986–2008 and 2009–2022. We take the cut-off point in 2009 to see the effect of the GFC on the publication themes in this area. We can see the impact in Figure 2.4: from 1968 to 2008, major publication themes included bank performance, bank privatization, and banks. However, after the crisis of 2008, the effects of GFC can be observed in the pattern of publication in the period 2009–2022, where the major theme of the publication is credit risk.

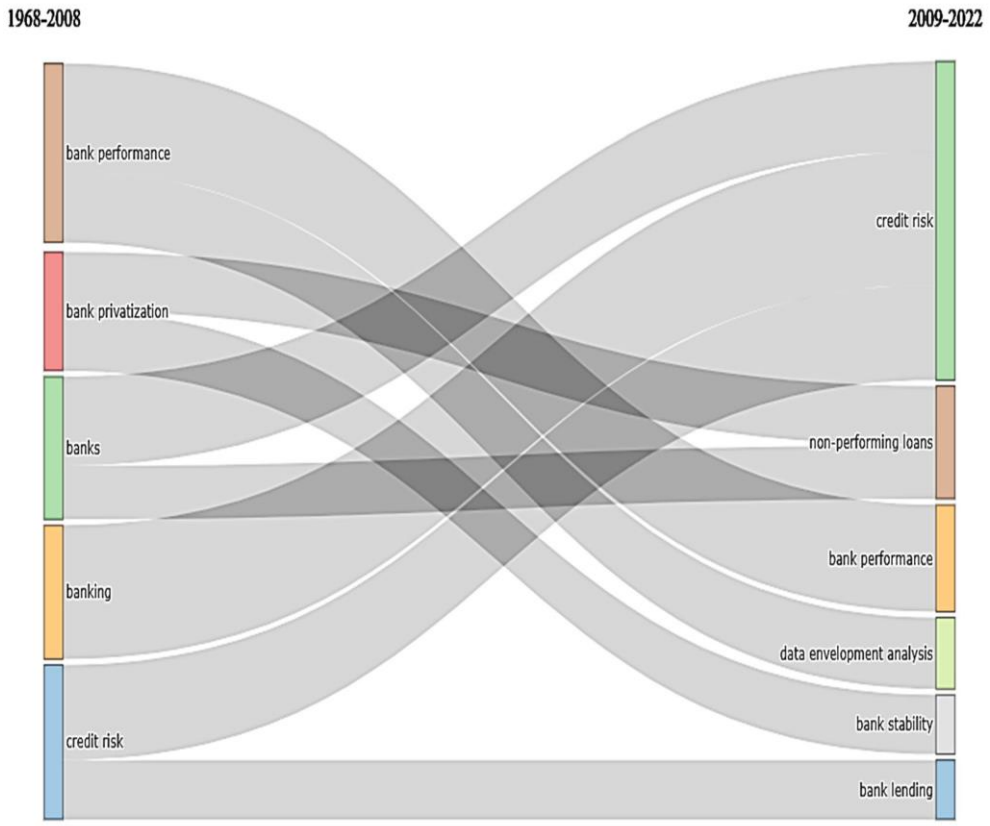

Figure 2.4 Thematic evolution over the years of publication

2.1.2.2 Analysis of author keywords

One of the important tools for analyzing publication trends is the analysis of author keywords. The author's keywords give a birds-eye view of the paper (Emich et al., 2020). Many authors have based their paper only on co-word analysis. Co-word analysis is a text-mining technique where co-occurrences of the key terms in publications are calculated, and then clusters are built based on themes and sub-themes(Sheikh et al., 2023). Table 2.1 lists the top 10 keywords from the Scopus and WoS database. The keyword analysis also resonates with the Thematic Evolution of "Credit Risk," which is the most used keyword, followed by "Bank Performance" and "NPLs."

S.no	Keyword	Frequency
1	Credit risk	103
2	Bank performance	62
3	Non-performing loans	61
4	Profitability	56
5	Bank profitability	50
6	Banks	47
7	Bank lending	39
8	Islamic banks	34
9	Banking	30
10	Performance	30

Table 2.1 Top 10 author keywords based on occurrences

2.1.3. Network Analysis

2.1.3.1 Thematic map

We can plot a thematic map by extending the keyword Co-occurrence network analysis. Words and their relationships are woven into thematic networks that display knowledge. A relative position is assigned to a thematic network using the centrality and the density within the global thematic network (according to their internal cohesion density)(Cobo et al., 2018).

Figure 2.5 shows a thematic map of the 50 most cited author keywords, categorizing the themes as motor themes, niche themes, basic themes, or emerging or declining themes based on the bibliometric analysis. According to this map, the most relevant and developed theme based on the keyword analysis relates to bank profitability and financial crisis as these are the motor themes, i.e., themes that are most developed and most relevant for the field of study(Cobo et al., 2018). This also resonates with our previous observations that the financial crisis and stability form the core of this research field following the 2008 financial crisis. The niche themes are the themes that are fewer in number and are very developed but are not of much relevance, monetary policy, bank lending channel, and bank efficiency. Data Envelopment Analysis, Islamic Banking, and Efficiency form part of emerging themes, and credit risk, NPA, bank performance, and profitability are the basic study themes in this field.

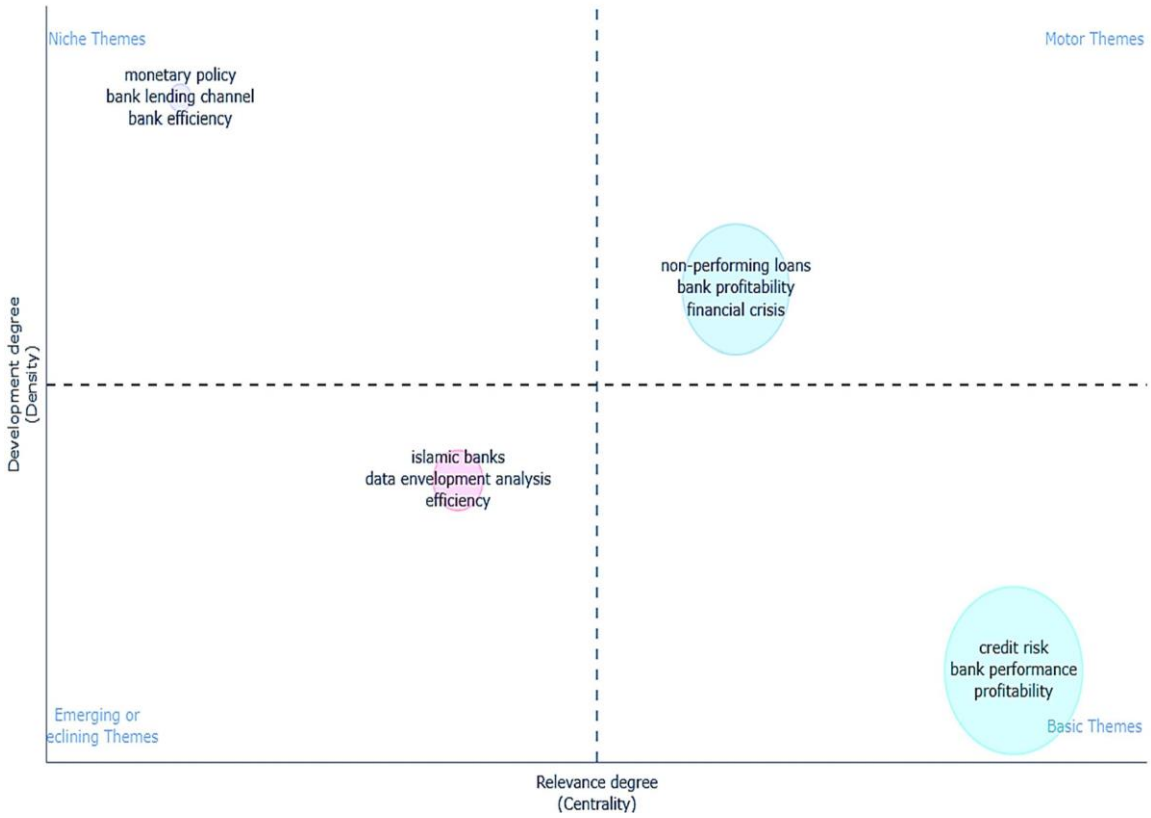

Figure 2.5 Thematic map of keywords

2.1.4 Thematic Analysis of Top 100 Papers

This section contains a detailed content analysis of the top 100 most cited papers, including 1 literature review and 99 empirical papers. In the content analysis section, we study the different themes of the papers, the methodology used by the authors, and the type of data used. We then analyze the findings of the paper. According to the thematic analysis of the top 100 papers, profitability determinants are the most dominant theme in this area. Out of a total of 100 papers, 38 are related to the bank profitability determinants based on different types of economies, such as emerging or developing or transient economies, and different types of banks, such as Islamic banks or conventional banks, and during different economic conditions such as during crisis or pre- or post-crisis period. The next dominant theme is bank lending

determinants, followed by efficiency measurement, mostly done using data envelopment analysis. Figure 2.6 provides insight into the different themes of the studies in this field.

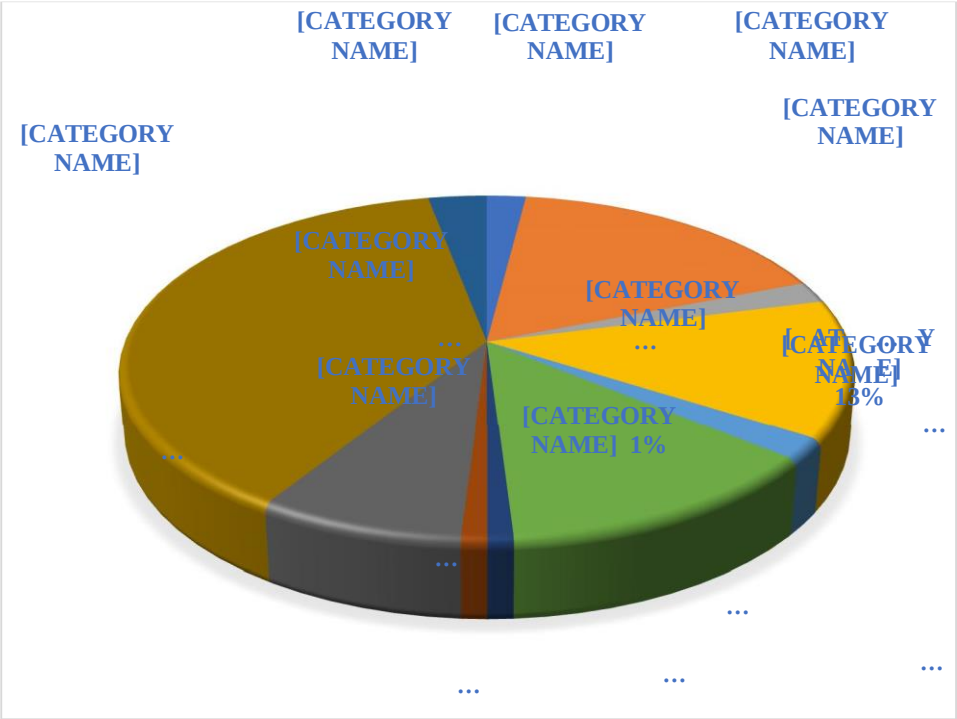

Figure 2.6 Thematic analysis of the Top 100 papers

2.2 Content Analysis

After exploring this stream of study using the bibliometric data, we move on to an in-depth understanding of the three basic tenets – Loan growth, Non-Performing Loans, and Bank profitability. Firstly, we understand these terms individually, and Then we understand their two-way relationship as a combination of two variables. Finally, we try to see what the existing literature says about the mutual relationship of all these three variables, i.e., the triad.

2.2.1 Loan Growth

A vital component of contemporary finance, bank loans give people and businesses the capital they need to operate, which is important to facilitate economic activity. To use this type of

financing, a borrower must get funds from a bank and usually agree to repay the loan amount plus interest over a predetermined period. Bank loan procedures and conditions can change depending on variables, including the borrower's creditworthiness, the loan's intended use, and the state of the market**.**

Loans can be made available as an open-ended line of credit with a predetermined maximum, or they can be made for a fixed, one-time amount. Bank loans are classified into consumer, commercial, secured, and unsecured.

Banks have lending targets – both internal and external. Internal targets are linked to the profit motive of the banks and are directed towards building a favourable credit mix. External targets are the targets the government or regulatory authorities give to meet social or economic welfare issues. The special thrust areas of government are often termed priority sectors, and lending in these areas is termed Priority sector lending.

Priority sector lending (PSL) norms have been implemented in Indian banks since 1968. Agriculture (and related activities), Micro and Small Enterprises (MSEs), Housing (mostly affordable housing up to a specified loan limit), Education, and Export Credit are the main sectors covered under PSL. In addition to this, PSL also covers loans made to State-sponsored organizations and Self-Help Groups that cater to underprivileged social groups. Due to their contribution to economic growth through employment creation and social redistribution, these industries are significant from both an economic and social standpoint.(Chavan & Gambacorta, 2019).

Banks have every reason to boost the volume of funds they lend as they are profit-driven enterprises. Bank loans can support firms operating and generating earnings that support economic growth (Dang, 2019a).

However, various factors influence banks' loan growth, both internal and external.

- Credit quality concerns – The pre-sanction appraisal and post-sanction follow-up of the loans are very important to ensure and maintain credit quality. The process of sanctioning a loan and following up on it till the time it is fully repaid requires expertise absence of which can create credit quality concerns. Hence, increasing a loan portfolio while maintaining the credit quality is tricky.
- Interest rate risk – Sudden changes in the short-term interest rates in the market can hamper the profit made by banks on the loans advanced as the interest earned is on a long-term interest basis; however, the interest borne by the bank is the market-regulated short-term interest rate. If the banks must mobilize money from the money market for lending, they must also be mindful of the interest rate risk.
- Market competition – Like any other business, competition also affects the banks' lending activities.
- Regulatory scrutiny – Regulators scrutinize the banks' lending activities as they deeply impact a country's economy.

This discussion on the bank's loan growth takes us to our next variable, i.e. NPLs.

2.2.2 Non-Performing Loans

Bank Loans that are not repaid within the stipulated period are called NPLs. International Monterey Fund (IMF) defined NPLs as if the interest accrued and the principal due for repayment remained unpaid for a minimum of 90 days, the loan will be termed NPL (Khan et al., 2020). Weak credit procedures, incompetent loan officials, high interest rate spreads, poor credit policy, and lack of post-sanction monitoring are the primary causes of high non-performing loans (NPLs). One important indicator of credit risk that impacts the nation's

banking sector is non-performing loans (NPLs). As NPLs impede the nation's economic growth by reducing credit creation, they might be used as a sign of impending banking crises. (Ivanovic, 2016). While a high number of NPLs indicates an unstable financial situation, a low level of NPLs demonstrates a robust monetary system in the nation. In the long run, the rising amount of NPLs will first have an impact on the commercial banks, and subsequently, it will have an impact on the financial standing of the national economy. (Souza & Feijó, 2011). The increasing amount of NPLs first affects the banking efficiency which might further result in banking crises (Vouldis & Louzis, 2018). NPLs hinder interest income, close opportunities for alternative investments, and cause liquidity problems in the financial system, all of which will contribute to a poor economy and bankruptcy. Therefore, to reduce the number of NPLs and achieve financial stability and economic objectives, it is imperative to understand the factors that influence NPLs (Stijepović, 2014). A high percentage of non-performing loans directly impacts the overall financial performance of banks. (Berger and DeYoung, 1997).

For Banks in India, RBI issues guidelines for Income Recognition and Asset classification from the year ending March 31, 2004, to promote greater openness and progress toward international best practices. Therefore, starting on March 31, 2004, a loan or advance falling behind on interest or a principal instalment for a term loan longer than ninety days is considered a non-performing asset.

(ii) In the case of an overdraft or cash credit (OD/CC) - the account stays "Out of order" for longer than ninety days.

(iii) In the case of bills that have been bought and discounted - the bill is still past due after more than ninety days.

(iv) The following guidelines apply to all direct agricultural advancements in the case of direct agricultural advances: a

a) A loan granted for short-duration crops will be treated as NPA if the instalment of principal or interest thereon remains overdue for two crop seasons.

b) A loan granted for long-duration crops will be treated as NPA if the instalment of principal or interest thereon remains overdue for one crop season

** Any amount due to the bank under any credit facility becomes overdue if not paid by the due date fixed by the bank***.** (Master Circular on Income Recognition, Asset Classification, Provisioning & Other Related Matters - UCBs, 2009)

After understanding the concept of NPLs – meaning and its nature we move forward to our next variable, bank profitability, so that we can move forward in our research to better understand the interaction of these three variables.

2.2.3 Profitability

Profitability is the end goal of any business activity; banks are no different. Be it public sector banks or private sector banks, both strive to maximize profits. Profitability is necessary for the shareholders who gain from it and is a very important indicator of the bank's health. A sound bank must be making profits, which also impacts the bank ratings that are awarded based on many aspects of bank performance, and profitability is one of the metrics. These bank ratings are responsible for helping the banks get funds from shareholders. Hence, it is necessary to understand what bank profitability is made of and what its determinants are.

The major portion of a bank's profits is the difference between the interest earned from the borrowers on the amount advanced to them and the interest paid to the depositors on

the amount they deposited in deposit accounts. Banks also earn income from other fee or commission-based activities, termed non-interest income.

The bank's profitability is determined by several factors, which can be classified into internal and external factors. The internal factors are the bank-specific factors, and the external factors are the industry-specific and the macroeconomic factors. Operating expenses, financial risk, capital holdings, and size are among the bank-specific determinants of profitability. The second set of determinants describes industry-structure characteristics that impact bank earnings but are not directly related to managerial actions, such as industry concentration. The third set of determinants concerns the macroeconomic context in which the banking system works. (O'Connell, 2023). Internal factors are the ones that are controlled or governed by management decision-making. The other two factors are the ones that management has to pay heed to while making any decision related to the policy or operations of the bank.

Drivers of profitability are many, such as boosting Net Interest Margin (NIM) and boosting the fee-based or commission-based income, which can increase the income and thus increase the profitability. Another perspective is cost or expenditure management. Operational or cost efficiency also increases profitability and is very important in banks' scenarios. NPA management plays a crucial role in increasing profitability. NPAs are a huge burden on a bank's profits as they are blocked assets that do not earn any interest and have to be provisioned for at very high rates, directly reducing the banks' profitability.

2.2.4 Loan Growth and Profitability

Lending is a risky business. The money advanced carries default risk. Rather than contributing to the bank's interest income, it may burden the bank's profitability by increasing

provisions (Almaqtari et al., 2018) and exploring the variables that determine the profitability of Indian banks. The study includes various bank-specific and macroeconomic variables; amongst these, the loan ratio of the bank does not show any significant impact on profitability (Sufian and Habibullah, 2009a, 2009b). These two studies are based on finding the determinants of profitability: one is based on a developing nation, Bangladesh and the other on a Southeast Asian economy—China. The loan intensity or loan growth results in both studies are different. In the case of Bangladesh, bank lending is positively related to profitability; however, this is not the case for the Chinese banking sector, where bank lending is inversely related to profitability. The differences in the results of the different studies make us understand that the impact of bank lending on a bank's profitability changes with the change in the study period, the region, and the nature of the economy (Adelopo et al., 2018a). (Adeolu, 2014)in their study based in Nigeria, while studying for 15 years, they found no significant impact on bank loans and profitability. Caporale et al. (2014) examine the factors that influence banks' profitability, paying particular attention to the periods before, during, and after the financial crisis, and conclude that bank lending negatively impacts the profitability during the crisis and after the crisis period, thus examining how the economic cycle affects the factors that determine profitability. The pattern of lending and the NPLs in the banking industry worldwide can be mapped with certain events in the nation's economic cycle. Bank lending increases in times of boom in the economy to meet the demand for credit, followed by an increase in the NPLS during the economic downturn. The reason for the same has been attributed to the lowering of the credit screening standards during the boom and the decreased disposable income to repay loans because of the recession. The study proposed that bank lending has a non-linear relationship with profitability.

Author & year	Country	Sample size	Methodology	Independent variables
(Adeolu, 2014)	MENA	30	Generalized method of moments	capital, ratio of cost and income, size, off-balance sheet exposure, credit, liquidity and interest rate risk, age, mergers and acquisitions, ownership, real GDP growth, inflation expectation
(Almaqtari et al., 2019a)	India	69	Random effects and fixed effects regression	Size, network of branches, Asset management operational efficiency, leverage ratio, inflation rate, exchange rate, interest rate, and demonization
(Sufian & Habibullah, 2009d)	Bangladesh	37	Multivariate regression analysis	Loan intensity, credit risk and cost, non-interest income, bank size, inflation, GDP
(Sufian & Habibullah, 2009b)	China	220	Multivariate regression	ROA, loans/TA, LNTA, LLP/TL, Non-interest Income ratio, non-interest expense ratio, Inflation, money supply, GDP
(Adelopo et al., 2018b)	Africa	123	Fixed effects regression	Size, liquidity, asset quality, market power, GDP, inflation, cost management, capital strength
(Caporale et al., 2014a)	Italy	17	Regression	Total bad loans, total loans, industrial production, unemployment rate, CPI, retail sales, housing price index, housing price index, Euribor interest rate (3 months), IRS 10 years, interest rate slope (3M – 10Y), FTSE MIB index

Table 2.2 Some Studies that link Loan Growth with Profitability

2.2.5 Loan Growth and Non-Performing Loans

Many existing studies have based their research on examining the impact of bank lending on NPLs. In their study on abnormal loan growth and loan losses, Foos *et al*. (2010) investigated the effect of a previous period's excessive loan growth on NPLs and found that, with a lag of two to four years, past excessive growth has a positive and highly significant impact on future loan losses. They also studied the two-way relationship between bank

lending and NPLs and found that increasing NPLs negatively impacts bank lending. The study on Japanese banks in the pre- and post-crisis periods found that bank lending has a positive relationship with the NPLs, but post-crisis period bank lending negatively impacts the NPLs (Vithessonthi, 2016a). We can conclude that loans are necessary for a crisis-stricken economy to recover. (Cuccinelli, 2016a) specifically studies the speed of lending and its relationship with asset quality over some time. According to the article, most banks that experienced faster growth before the financial crisis also had higher rates of non-performing loan growth and higher rates of lending activity decline during the crisis**.** Alhassan et al. (2014) also studied the asset quality using the NPLs in the crisis period in Ghana and concluded that bank lending positively impacts the NPLs, i.e., as the loans increase, NPLs also increase. Assuming a linear negative relationship between NPLs and loan growth does not fully justify the nature of the relationship between these two aspects (Boussaada et al., 2020). Their study based on banks of the MENA region studies the threshold effect of liquidity risk on NPLs. Fahlenbrach et al. (2018)**,** in their study based on U.S. banks, found that the banks with excessive lending show poor returns, and the reason behind this is attributed to the prediction of risky lending by the bank. A Vietnamese studying the long-term impact of loan growth on NPLs found that lending increases loan loss provisions for 2 to 3 subsequent years (Dang, 2019a) **(**Laidroo & Männasoo, 2013); this study also finds that an excessive amount of credit availability forecasted a rise in bank loan loss reserves one year ahead of schedule despite accounting for actual loan portfolio growth and GDP growth in the nation.

Author & year	Country	Sample size	Methodology	Independent Variable
(Cucinelli, 2016b)	Italy	513	Least absolute deviation, OLS, fixed effect regression	NPA/gross advances, bad debts provision/gross advances, addition to NPA/gross advances, ratio of advances and deposit, rate of growth of deposit, equity to total asset, unemployment, GDP, and inflation.
(Dang, 2019b)	Vietnam	31	Static and dynamic panel regression	Capital, size
(Foos et al., 2010b)	Global	16000	OLS regressions and a dynamic two-step system generalized method of moments panel estimator	Capitalization, abnormal growth of advances, cumulative loan losses.
(Laidroo & Männasoo, 2013)	Central and Eastern Europe	337	Generalized method of moments	The ratio of bad debts provision to gross advances and assets, real loan growth, real GDP growth, Pre-impairment operating profit to assets, unused committed credit lines to total loans
(Vithessonthi, 2016b)	Japan	82	OLS	NPLTA, Bank credit growth, Bank size, Bank credit. Capitalization ratio, liquidity ratio, revenue diversification ratio
(Alhassan et al., 2014b)	Ghana	25	GMM	NPL ratio, substandard loan ratio, doubtful loan ratio, LLR, HHI, loan growth, bank size, Net interest income ratio, Income diversification, Non-Interest Income/Total Income, GDP, inflation, real exchange rate

(Boussaada et al., 2022)	MENA	72	Panel threshold regression	NPLs, ownership concentration, board size, liquidity risk, ROA, size, capital, crisis, GDPG, INF
(Fahlenbrach et al., 2018b)	U.S.A	200	regression	Loan growth, ROA, ROE, Loan loss provision, long-term growth forecast, Loans to assets, Bank age, Tier 1 capital, Equity to asset ratio.

Table 2.3 Some Studies that Link Loan Growth with Non-Performing Loans

2.2.6 Non-performing loans and bank profitability

Researchers have also studied the relationship between NPLs and bank profitability in various regions of their research. In their study, Kadioglu et al. (2017) have incorporated the deteriorating quality of the assets and bad loans. It is found that non-performing loans and bank profitability, as determined by return on equity and return on asset, have a substantial, inverse relationship. Return on equity and return on asset are negatively correlated with greater non-performing loan levels; conversely, higher non-performing loan levels are positively correlated with lower asset quality and higher return on equity and return on asset. (Laryea et al., 2016a) also base their study on the procyclicality of the NPLs.They find that when the economy is doing well, bank customers can make enough money from their investments to repay their loans. This can reduce the number of non-performing loans (NPLs). They also find data suggesting NPLs have a negative impact on ROE and ROA, defying the theory that greater risks yield higher returns. It is suggested that banks build their credit advancement policies primarily on economic performance, since non-performing loans (NPLs) tend to increase in weak economic times and decrease in strong economic timesNPLs impact profitability by increasing the amount of provisioning to be done. Also, the capital

requirements of the banks increase as the assets become riskier, which entails a considerable capital cost to the banks, thus reducing profitability.(Mostak Ahamed, 2017a), in their study based on the Indian banking sector, keeping in mind the costs of higher NPLs, which bring down the profitability, suggest that the banks with lower asset quality must consider diversifying and concentrating more on the non-interest income through fee-based or commission-based activities to stabilize or boost their profits. However, a study on loan diversification to reduce NPLs and boost profitability in an emerging economy, Ghana, reveals different results.(Adzobu et al., 2017a) in their study emphasize the necessity for banks operating in emerging economies to carefully assess how diversifying into new industries may influence both the incentives and efficacy of their supervision.

The bank's experience and resources for loan screening and monitoring significantly impact credit risk. Therefore, banks must continue to invest in their risk management procedures to ensure effective and efficient monitoring of their loan portfolios and improve credit quality. Since research indicates that more diversification lowers bank profitability and raises credit risk, banks in emerging nations may also want to consider the alternative approach of focusing their lending in a small number of industries.(Alqahtani et al., 2022) Propose a non-linear relationship between NPLs and profitability using a global sample. Using the GMM and quantile regression, they find that NPLs first reduce when the profit increases but increase as the profit level increases in a bank. This can be attributed to many underlying facts, which will be discussed in the theoretical framework of our study.

Author & year	Country	Sample size	Methodology	Independent variables
(Adzobu et al., 2017b)	Ghana	30	Generalized method of moments	Diversification, capital adequacy, liquidity, bank size, GDP growth
(Akram & Rahman, 2018)	Pakistan	9	Multiple regression and descriptive analysis	Growth of total assets, gross loans, NPA/gross advances, bad debts reserve, LLR/gross advances, LLP/NII, LLR/NPA, ratio of net charge-off with average NPAs and gross loans, unreserved impaired loans/equity, bank's credit risk management
(Boahene et al., 2012)	Ghana	6	Regression, variance inflation factor (VIF) analysis	Rate of non-performing loans, the net rate of charge-off, percentage of profit to total loans, ROE, size, growth, leverage
(Duho et al., 2020)	Ghana	32	DEA and accounting ratios	Provision/gross advances, securities/assets, liquid assets/total deposits,15% of 3 years average income to equity, ratio of equity to total assets, Z-score5 (ROA þ Asset/Equity)/σ(ROA), age, size.
(Kwan & Eisenbeis, 1997b)	US	352	Simultaneous equation framework	Doubtful debts to gross advances, difference between assets and liabilities, equity to total assets, real estate loans, commercial loans, and industrial loan to gross advances, growth of advances, square of the growth rate of gross advances, ROA, assets;
(Alqahtani et al., 2021)	Global	2943	Generalized method of moments (GMM), Quantile regression	NPA to gross advances, size, the ratio of cost and income, ratio of equity and assets, ratio of gross advances and deposits, customer deposit to liabilities, total assets, GDP, inflation, unemployment

(Kadioglu et al., 2017)	Turkey	55	Panel regression	NPA /total assets, provision for NPA /gross advances, total equity/total assets
(Laryea et al., 2016b)	Ghana	22	Fixed effect panel model	Non-performing loans, interest rate, credit advanced, CAR, size, asset concentration of banking industry, inflation, GDP
(Mostak Ahamed, 2017b)	India	107	Panel regression	Bad debts provision, ratio of gross loans loan to total assets

Table 2.4 Some Studies that Link Non-Performing Loans with Profitability

2.2.7 The triad

After discussing these variables in isolation and combining a set of two variables, we discuss the triad. The main objective of our study is to develop a framework that interrelates the three most important variables of banking – Loan growth, NPLs, and profitability. As stated in section 2.1 on conducting a WoS and Scopus search of these three variables, we found only 27 studies. When we go through the abstract of these 27 studies, we find that none have these three variables to see the interrelationship. They study the relationship between these variables in sets of two and do not try to connect the dots to understand their mutual relationship. We review some studies that discuss these three variables and try to build our study further.

(Twum et al., 2022) in a recent study based on China, the fastest-growing economy in the world, the research examined the connection between credit risk and company performance using panel data for 28 listed commercial banks in China between 1990 and 2020. Non-performing loans, return on equity, loan loss provision, and bank profitability were negatively and statistically significantly correlated. Conversely, a positive and statistically

significant link was found between bank performance and the capital adequacy ratio. On the other hand, credit growth showed an insignificant favourable correlation with bank performance. The link between the loan growth and NPLs was missing in the study. (Thornton & Di Tommaso, 2021) study based on European banks examine if, in a panel of up to 521 banks from 28 European nations, the impact of non-performing loans (NPLs) on bank lending growth varies according to the amount of bank capital and profitability. The study concludes that the availability of bank credit is significantly impacted positively by the connection between non-performing loans (NPLs) and bank capital, as well as NPLs and profitability. Therefore, the extent to which NPLs obstruct the transmission mechanism of monetary policy largely depends on banks' profitability and adequate capitalization. If policy initiatives are to successfully lower NPLs and maintain bank credit, they must safeguard bank capital and profitability.

This study checks the impact of capital and profitability on the relationship between loan growth and NPLs. Another study based on European banks studies the trade-off between excessive lending and performance in terms of risk premium using the risk management hypothesis and moral hazard hypotheses from the existing banking literature. They find that Banks normally trade off the loan growth rate with the risk in their portfolio. But their actions may cause others to "gamble to survive" or "search for yield." The risk management hypothesis is valid for large banks' years following the 2007–2008 financial crisis. On the other hand, the pre-crisis period and small banks support the moral hazard hypothesis, which holds that banks lend excessively to relatively risky borrowers who pay greater premiums but increase their credit risk.

All these studies and similar studies focus on our variables under study, but none try to form a framework to explain their interrelationship that explains their behaviour. This brings us to explain the research gap in this field of study.

2.3 Research Gap

The literature review leads us to finding a research gap in the existing literature. The motivation of our study is to understand the impact of loan growth on NPLs and profitability and the mediating role of NPLs in this relationship. For this, we must develop a framework that connects all these three variables and explain their relationship and mutual relationship.

2.3.1 Literature gap

As mentioned in section 2.1, we find only 27 studies that study the three variables together. Also, in section 2.2.7, we see in-depth these studies to find that none talk about these three variables concerning each other, and none talk about the mediating role of NPLs in the relationship between loan growth and profitability. Some studies discuss the trade-off between loan growth and risk but segregate it into pre-crisis and post-crisis periods. Thus, we see a literature gap about these three variables' interrelationships to understand these behaviours in the Indian banking industry.

2.3.2 Meta-Analysis

2.3.2.1 Summary of findings of the existing studies

This study aims to understand how Loan growth impacts NPLs and profitability and the impact of NPLs on the bank's profitability. From the literature review of the articles, we summarise the relationship between our variables of interest in Table 2.5. The three main

relationships of our interest are bank lending impact on NPL, bank lending impact on profitability, and NPL impact on profitability.

	Positive	Negative	Non - Significant	Inverse U shape
Loan growth impact on NPL	9	3		
Loan growth impact on profitability	7	11	2	1
NPLs' Impact on Profitability	4	16	1	

Table 2.5 Summary of the findings

The heterogeneity in the study's findings motivates us to do a meta-analysis of the results of the articles reviewed, which can help further the study's aim and strengthen its model.

We used meta-analytical techniques to synthesize the empirical data for the association between loan growth, NPLs, and profitability. The primary goal of this meta-analysis is to appropriately aggregate the "effect sizes" gathered from a sample of studies. The magnitude and direction of a correlation coefficient concerning a relationship of interest are examples of effect size (Hunter & Schmidt, 2004). According to several academics (Dalton & Dalton, 2005), it has become more common in management and finance research to quantitatively integrate data from numerous studies to determine whether there are consistent links among a group of variables.

For our study, we chose the top 200 papers based on citations. From these 200 papers, we select the articles that study the two-way combination of our variables of interest and extract the correlation coefficients and the sample sizes. As many studies do not report correlation coefficients in their publication, we have to limit ourselves to the studies that have

reported the same. We find a sample of 18,21, and 25 studies for studying the relationship between bank lending and profitability, Bank lending and NPLs, and NPLs and profitability, respectively.

We use the meta essentials correlation coefficient meta-analysis worksheet to perform the meta-analysis(Suurmond et al., 2017). As we are using the Pearson correlation coefficients for meta-analysis, it needs to be transformed using the Fisher r-to-z transformed correlation coefficient, and then the random effects model is fitted with a 95% significance level. The level of heterogeneity is measured using the Q test for heterogeneity. The I^2 value shows the level of heterogeneity in the effect sizes. An I^2 value of 0-50% is considered homogenous, 0-75% is moderately heterogeneous, and above 75% heterogeneity is considered a high heterogeneity level(Harris et al., 2008; Higgins & Thompson, 2004).In case of high heterogeneity, a random effects model is fitted. Random-effects meta-analysis model assumes that the effects being estimated in the different studies are not identical but follow some distribution[1]. A funnel plot is used to assess publication bias due to a small sample size. Eggers test (Egger et al., 1997) and Begg and Mazumdar (Begg & Mazumdar, 1994) test is carried out to assess any funnel plot asymmetry.

2.3.2.2 Meta-Analysis Results and Discussion

2.3.2.2.1 Loan growth impact on profitability

A total of 18 studies were chosen based on the appropriateness and availability of data required for meta-analysis. Table 2.5 shows the list of studies and the correlation coefficients ranging from -0.3316 to 0.3541; 61% show a negative correlation between these two variables. The heterogeneity test shows that the results are heterogeneous as the I^2 is 95.67 %, which depicts high-level heterogeneity. The student's residuals show no outliers in the group, and Cook's distance reveals none of the studies to be overly influential.

Figure 2.7 is a forest plot of the effect sizes of the 18 studies. The black box's width depicts the study's weight, and the horizontal line's length depicts the lower and the upper CI limits. The last row, i.e. the 19th row, shows the pooled ES of all 18 studies using the random effects model. The figures show that there is heterogeneity in the results of the various. However, neither the Eggers test (p= 0.87) nor the Begg and Mazumdar rank correlation (p= 0.8201) shows publication bias.

Study	**Study Name**	**Correlation**	**CI Lower limit**	**CI Upper limit**	**Weight**
A	"Credit risk securitization and bank soundness in Europe"	-0.05	-0.14	0.05	5.61%
B	"Factors influencing bank profitability in a developing economy empirical evidence from Malaysia"	-0.32	-0.51	-0.14	4.60%
C	"The effect of the political connections of government bank CEOs on bank performance during the financial crisis"	0.12	0.04	0.2	5.71%
D	"The determinants of liquidity of Indian listed commercial banks a panel data approach"	0.05	-0.05	0.15	5.54%
E	"Commercial bank ownership and performance in China"	-0.12	-0.2	-0.04	5.73%
F	"The effects of loan growth on bank performance evidence from Vietnam"	0.35	0.22	0.49	5.17%
G	"Is there a bright side to government banks' evidence from the global financial crisis"	0.09	0.05	0.13	6.00%
H	'The simultaneous estimation of credit growth valuation and stability of the Gulf Cooperation Council banking industry"	-0.02	-0.11	0.07	5.69%
I	"The effect of credit risk liquidity risk and bank capital on bank profitability evidence from an emerging market"	-0.05	-0.23	0.13	4.62%
J	"Competition financial intermediation and riskiness of banks evidence from the Asia-Pacific region"	-0.13	-0.17	-0.1	6.01%

K	"The determinants of profitability of Indian commercial banks a panel data approach"	-0.30	-0.37	-0.22	5.78%
L	"Deflation bank credit growth and non-performing loans evidence from Japan"	0.07	0.02	0.12	5.92%
M	"Should Bankers be concerned with intellectual capital: a study of the Thai banking sector."	-0.32	-0.38	-0.27	5.91%
N	"Bank lending behaviour in emerging markets"	0.09	-0.03	0.2	5.40%
O	"Bank stability in the case of Nordic and non-nordic banks in Latvia"	-0.21	-0.33	-0.09	5.32%
P	"Determinants of bank profitability before during and after the financial crisis"	0.09	0.03	0.21	5.95%
Q	"Bank specific and macroeconomic determinants of profitability of Indian commercial banks a panel data approach"	-0.33	-0.41	-0.26	5.78%
R	"The impact of bank capital bank liquidity and credit risk on profitability in the post-crisis period a comparative study of us and Asia"	-0.04	-0.17	0.09	5.27%
	Pooled ES	-0.06	-0.14	0.03	100%
	Heterogeneity Chi-squared	392.43			
	I^2	95.67%			

Table 2.6 Meta-Analysis results and effect sizes; Loan growth impact on profitability

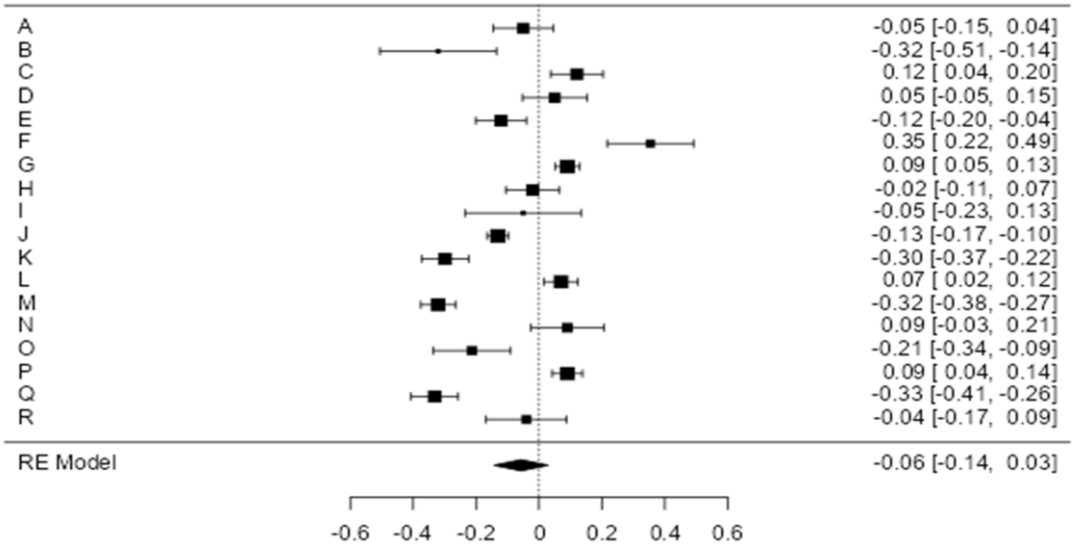

Figure 2.7 Forest plot bank lending impact on profitability

2.3.2.2.2 Loan Growth Impact on NPLs

We have used 21 studies to study this relationship, details of which are in Table 2.7, chosen based on data availability and relevance. The transformed correlation coefficients range from -0.3316 to 0.4118. 62% of the estimates are negative. The pooled ES is -0.06 with CI- -0.013 to 0.02. The results are heterogeneous, with an I^2 value = 96.09%. The student's residuals show no outliers in the group, and Cook's distance reveals none of the studies to be overly influential. Figure 2.8 is the forest plot depicting the individual and pooled effect sizes of the 21 studies, and Neither the Eggers test (p= 0.6449) nor the Begg Mazumdar test (p= 0.4501) shows any funnel plot asymmetry.

Study	Study Name	Correlation	CI Lower limit	CI Upper limit	Weight
A	"Corruption and banking stability evidence from emerging economies"	0.41	0.34	0.48	5.20%
B	"Determinants of bank profitability in developing economies empirical evidence from the South Asian banking sectors."	-0.16	-0.24	-0.08	5.10%

C	"Factors influencing bank profitability in a developing economy empirical evidence from Malaysia"	-0.11	-0.30	0.07	3.91%
D	"The effect of the political connections of government bank CEOs on bank performance during the financial crisis"	0.00	-0.08	0.08	5.07%
E	"Commercial bank ownership and performance in China"	0.15	0.07	0.23	5.09%
F	"The effects of loan growth on bank performance evidence from Vietnam"	0.00	-0.14	0.14	4.49%
G	"Loan growth and bank solvency evidence from the Pakistani banking sector"	-0.23	-0.36	-0.11	4.63%
H	"Discretionary provisioning practices among Western European banks"	-0.33	-0.42	-0.25	5.06%
I	"Is there a bright side to government banks' evidence from the global financial crisis?"	-0.17	-0.21	-0.13	5.39%
J	"The simultaneous estimation of credit growth valuation and stability of the Gulf Cooperation Council banking industry"	-0.02	-0.11	0.07	5.05%
K	"The effect of credit risk liquidity risk and bank capital on bank profitability evidence from an emerging market"	0.01	-0.17	0.19	3.93%
L	"The implications of credit risk modelling for banks' loan loss provisions and loan origination procyclicality"	-0.01	-0.03	0.01	5.46%
M	"What drives non-performing loans evidence from emerging and advanced economies during pre and post-global financial crisis."	-0.17	-0.33	-0.01	4.23%
N	"Bank loan loss provisions investor protection and the macroeconomy"	-0.16	-0.20	-0.12	5.38%
O	"Competition financial intermediation and riskiness of banks evidence from the Asia-Pacific region"	-0.24	-0.28	-0.21	5.40%
P	"Determinants of bank profitability in a developing economy empirical evidence from Bangladesh"	0.20	0.03	0.38	4.04%
Q	"Deflation bank credit growth and non-performing loans evidence from Japan"	-0.12	-0.17	-0.07	5.30%
R	"Bank lending behaviour in emerging markets"	-0.02	-0.14	0.10	4.73%
S	"Determinants of bank profitability in a	-0.26	-0.44	-0.07	3.96%

	developing economy empirical evidence from the China banking sector"				
T	"Determinants of bank profitability in Tunisia: does corruption matter."	0.10	-0.08	0.28	3.96%
U	"The impact of bank capital bank liquidity and credit risk on profitability in the post-crisis period a comparative study of us and Asia"	0.01	-0.12	0.14	4.60%
Pooled ES		-0.06	-0.013	0.02	100%
Heterogeneity Chi-squared		511.14			
I^2		96.09 %			

Table 2.7 Meta-Analysis results and effect sizes – Loan growth impact on NPLs

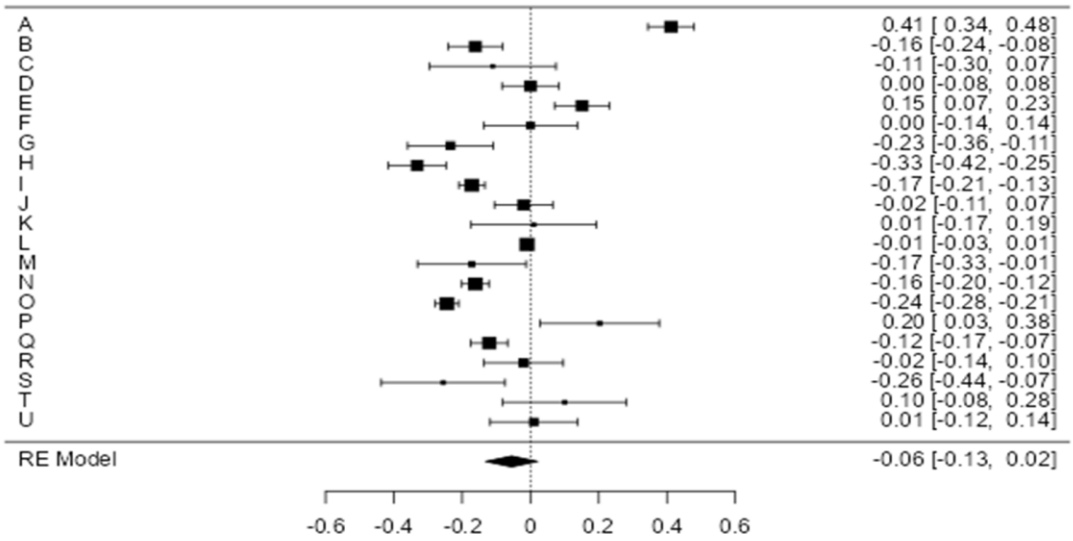

Figure 2.8 Forest Plot -Loan Growth Impact on NPLs

2.3.2.2.2 NPLs' impact on profitability

Table 2.8 shows the 25 studies are included in the analysis. The I^2 value = 99.61% shows a high level of heterogeneity. One study (O), which may be an outlier in the context of this

model, had a value larger than 3.0902 when the students' residuals were examined. Cook's distances suggest that one study (O) may have excessive influence.

Figure 2.9 is a forest plot of the individual and pooled effect sizes. Neither the Begg Mazumdar nor Eggers tests indicated any funnel plot asymmetry (p = 0.2332 and p = 0.4536, respectively).

Study	**Study Name**	**Correlation**	**CI Lower limit**	**CI Upper limit**	**Weight**
A	"Factors influencing bank profitability in a developing economy empirical evidence from Malaysia"	-0.52	-0.65	-0.39	3.91%
B	"Does bank competition matter for performance and risktaking empirical evidence from BRICS countries"	0.24	0.19	0.29	4.09%
C	"The effect of the political connections of government bank CEOs on bank performance during the financial crisis"	-0.08	-0.16	0.00	4.07%
D	"Unveiling the energy-saving role of banking performance in sub-Sahara Africa"	-0.10	-0.20	0.00	4.05%
E	"Determinants of banks' profitability role of poor asset quality in Asia"	-0.38	-0.40	-0.36	4.11%
F	"Commercial bank ownership and performance in China"	-0.42	-0.49	-0.35	4.07%
G	"Islamic banks and political risk international evidence"	0.28	0.27	0.29	4.11%
H	The Effects of loan growth on bank performance evidence from Vietnam	-0.02	-0.16	0.12	4.00%
I	"Determinants of non-performing loans in the banking sector in small developing island states a study of Fiji"	-0.52	-0.67	-0.37	3.88%
J	"Is there a bright side to government banks' evidence from the global financial crisis?"	-0.54	-0.57	-0.51	4.10%
K	"The simultaneous estimation of credit growth valuation and stability of the Gulf Cooperation Council banking industry"	-0.16	-0.24	-0.08	4.07%

L	"The conditioning role of performance on the bank risktaking channel of monetary policy evidence from a multiple tool regime"	-0.28	-0.38	-0.18	4.04%
M	"The effect of credit risk liquidity risk and bank capital on bank profitability evidence from an emerging market"	0.34	0.18	0.50	3.92%
N	"A multivariate analysis of the financial characteristics of foreign and domestic banks in the UK"	0.02	-0.11	0.15	4.01%
O	"CEO characteristics and bank performance evidence from India"	0.80	0.76	0.84	4.03%
P	"The risk-return trade-off of liquidity positions evidenced by the Vietnamese banking system"	-0.06	-0.17	0.05	4.04%
Q	"Non-performing loans and bank profitability evidence from an emerging market"	-0.13	-0.30	0.04	3.94%
R	"Competition financial intermediation and riskiness of banks evidence from the AsiaPacific region"	0.36	0.33	0.39	4.10%
S	"Liquidity risk and performance of banking system"	-0.24	-0.40	-0.08	3.94%
T	"Deflation bank credit growth and non-performing loans evidence from Japan"	-0.43	-0.47	-0.39	4.09%
U	"Should Bankers be concerned with intellectual capital: a study of the Thai banking sector."	-0.15	-0.20	-0.10	4.09%
V	"Bank lending behaviour in emerging markets"	0.00	-0.12	0.12	4.03%
W	"Determinants of bank profitability evidence from Vietnam"	-0.30	-0.40	-0.20	4.03%
X	"Comparative study of the performance of Islamic and conventional banks in the case of Malaysia"	-0.38	-0.73	0.03	3.24%
Y	"The impact of bank capital bank liquidity and credit risk on profitability in the post-crisis period a comparative study of us and Asia"	0.02	-0.11	0.15	4.01%
Pooled ES		-0.10	-0.23	0.03	100%
Heterogeneity Chi-squared		6212.16			
I^2		99.61%			

Table 2.8 Meta-analysis results and effect sizes, NPLs and profitability

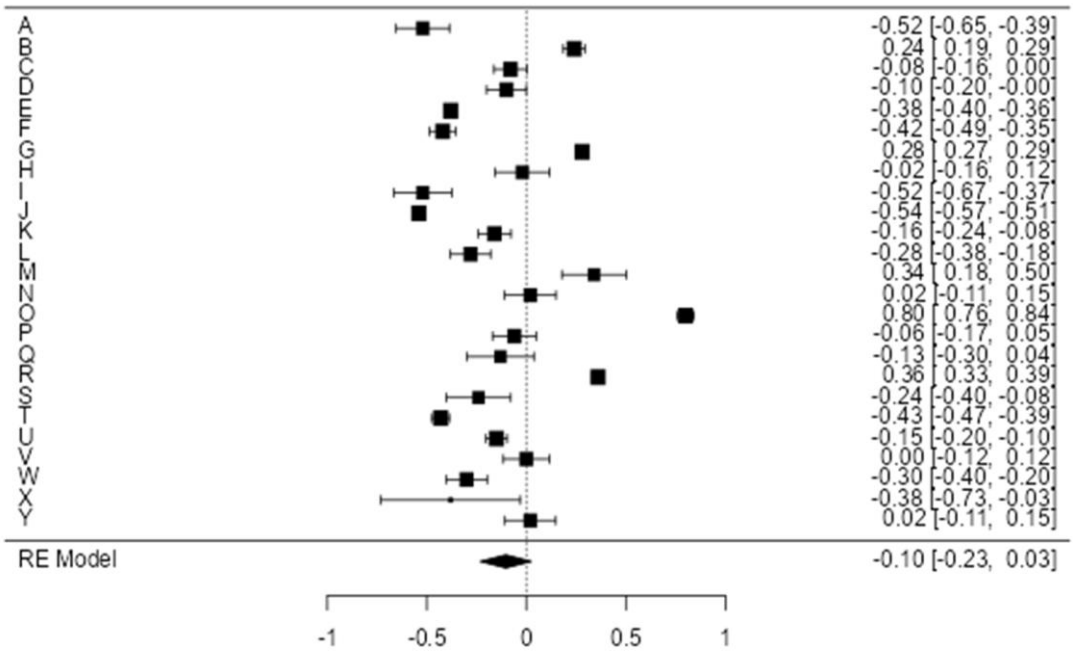

Figure 2.9 Forest plot, NPLs impact on profitability

The meta-analysis of the relationship between three variables establishes a negative relationship between loan growth and profitability, loan growth and NPLs, and NPLs and profitability. As per the results from the meta-analysis, the increase in loan growth reduces profitability and NPLs, and an increase in NPLs has a negative impact on profitability. However, high heterogeneity in the effect sizes of the selected studies makes it difficult to summarise the findings.

The heterogeneity can be attributed to the geographical differences in the studies or different economic conditions (pre-crisis or post-crisis) studied. The interaction of one variable with another may change with the change in the economic conditions, i.e. bank lending in a recession may have a different impact on NPL than bank lending in the boom phase.

2.3.3 Identifying the Research Gap

Figure 2.10 shows a Venn diagram of the three key variables of our study. During the literature review, we found an extreme shortage of papers that study the interaction of these variables.

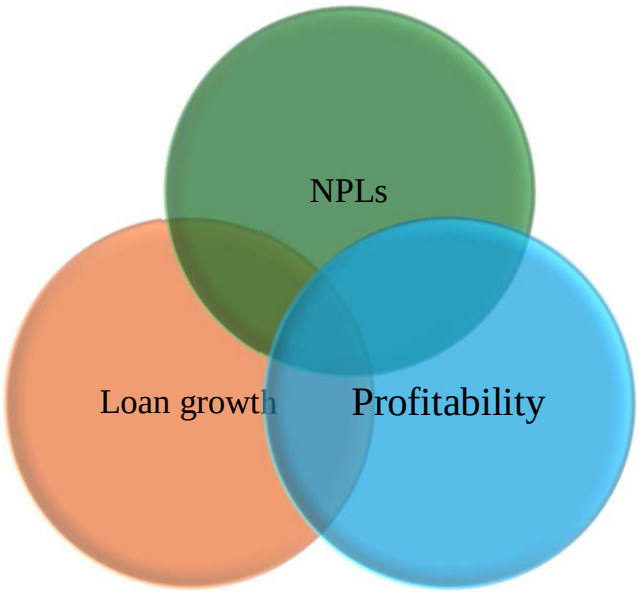

Figure 2.10 Venn diagram showing the impact of each variable on the other variable.

1. No literature studies the NPLs and Profitability as the Dependent variable, and the impact of loan growth is studied on both variables simultaneously.
2. No literature simultaneously studies the impact of NPLs on profitability and segregates the direct and indirect impact of loan growth on profitability by studying the mediation effect of the NPLs.

3. The heterogeneity in the results of existing studies shows that the impact of these variables changes with time and in different economic scenarios. Thus, it is necessary to summarise their behaviour which aids in long-term and all-encompassing policy making.

CHAPTER 3

THEORETICAL BACKGROUND & RESEARCH METHODOLOGY

3.1 Theoretical Background

The relationship between loan growth and profitability is a base for bank policy formulation. Maintaining asset quality is one of the main considerations while concentrating on loan growth. If the accelerated pace of lending is accompanied by a deteriorating asset quality, i.e., increasing NPLs then it does not serve any purpose. As we see in the existing literature, different researchers' views and empirical findings vary in this respect.

In our study, we build our hypotheses based on the four hypotheses (Berger et al., 1997) on problem loans and cost efficiency and link it with loan growth through the cyclic nature of economic activities.

3.1.1 The four hypotheses

As mentioned, the study titled "Problem Loans and Cost Efficiency in Commercial Banks" by (Berger et al., 1997) states four hypotheses to understand and empirically test the relationship between the cost efficiency and quality of loans in banks.

"Cost Efficiency" means efficiently using monetary or non-monetary resources to achieve more profits. A cost-efficient business is assumed to be a well-managed business. However, it is very important to realize that cost-efficiency should not be attained by cost-cutting in core areas of a business. For example, cost-cutting in product safety checks may lead to the production of low-quality products that may lead to a decline in sales.

Also, cost efficiency is linked with management expertise; if a business unit is operating cost-efficiently, it is assumed that its managers are also efficient and make good decisions for better working of the business.

The following four hypotheses are put forward by (Berger et al., 1997) -

1. Bad Luck - The Bad Luck hypothesis states that some external events that are not in the control of the management, such as a war situation in the country, lead to a rise in delinquencies. An increase in bad loans means more resources must be invested to recover the bad loans. This increase in expenditure decreases the cost efficiency for a longer period. Such external events can cause any organization to become inefficient in terms of costs. The Bad Luck hypothesis states that an increase in NPLs causes cost inefficiency, which can be attributed to the above-mentioned reasons. The NPLs leading to inefficiency is a hypothesis studied by many researchers (S. Ghosh & Sanyal, 2019; Salike & Ao, 2018) who have found a causal relationship between the NPLs and cost inefficiency.
2. Bad Management – After "Bad Luck" which assumes that the cause of rising NPLs is external and rising NPLs cause the cost inefficiency to rise. Under "Bad Management" hypothesis, it is assumed that if an organization is inefficient, it means that the management is not capable enough, and incapable managers will not be able to manage the loans efficiently, which will lead to a rise in NPLs.The "Bad Management" hypothesis states that the inefficiency causes the NPLs.The hypothesis assumes that the incapable managers are not skilled enough to a) undertake proper pre-appraisal related to the credit rating and collateral valuation, thus sanction loans with low ratings that have an inherent risk of turning bad, and b) they also do not possess skills for proper post sanction followup and monitoring of loans that start to

become out of order and thus more loans turn bad. The “Bad management” and “Bad Luck” hypotheses assume a negative relationship between cost efficiency and NPL but in a reverse sequence. This hypothesis brings into the picture a very important part: the skills required by the loan officials of the banks for sanctioning and managing loans. Despite being a routine task for the bank, loan growth requires special pre- and post-sanction follow-up skills.

3. Skimping - It states that the amount of resources invested in underwriting and monitoring the loans affects the cost efficiency and the asset quality. The Bank has a very important decision to make whether they want to invest more resources in the underwriting and monitoring of the loans to maintain the loan quality in the long run or save short-term costs to achieve more profits. Skimping on the resources invested in screening the applications, collateral values assessment, and monitoring of the loans may appear to be cost-efficient in the short run but it results in more delinquencies in the loan portfolio of the banks. Thus this hypothesis says that cost efficiency in one period causes problem loans later. The “Skimping Hypothesis” says that cost efficiency is positively related to problem loans.(Alsharif, 2021) tests the skimping hypotheses in Gulf Cooperation Council (GCC) countries and finds that it holds for the conventional banks in the GCC as opposed to the Islamic banks.(Sarkar et al., 2019) also confirm the presence of skimping hypotheses in the foreign banks of India, stating that the managers of foreign banks may have more incentive to report short-term profits.
4. Moral hazard – The next and last hypothesis is the moral hazard hypothesis, which links the skimping behaviour with the size of the banks. The smaller banks with less capital are more inclined towards taking risks as they have fewer stakes. They strive to

earn more profits by taking riskier loans at higher interest rates, as low-rated loan facilities attract higher interest rates. The " Moral hazard" hypothesis links the capital with the problem loans. Banks with larger capital have larger stakes at risk and, thus, are not involved in risky lending. (Abbas et al., 2021) studied the risk-taking behaviour of U.S commercial banks in the light of bank capital and found the moral hazard hypothesis to hold.

These four hypotheses form a basis of research for a wide range of banking literature. It tries to link the problem loans and cost efficiency through managerial decision-making. We use this study as a base for creating a theoretical framework for our study. We now link this with the cyclical nature of loans and problem loans by understanding the economic cycle.

3.1.2 The Procyclicality of Loans and NPLs

In the introduction section 1.3 we discussed the impact of the global economic scenario on the banking sector in India. We saw how the economic events in the U.S.A. had an impact on the Indian Banking Sector as well as the banks globally. Economic activities follow a cyclical pattern termed an economic or a business cycle. It is divided into four stages: Expansion, Boom, Recession, and Depression.

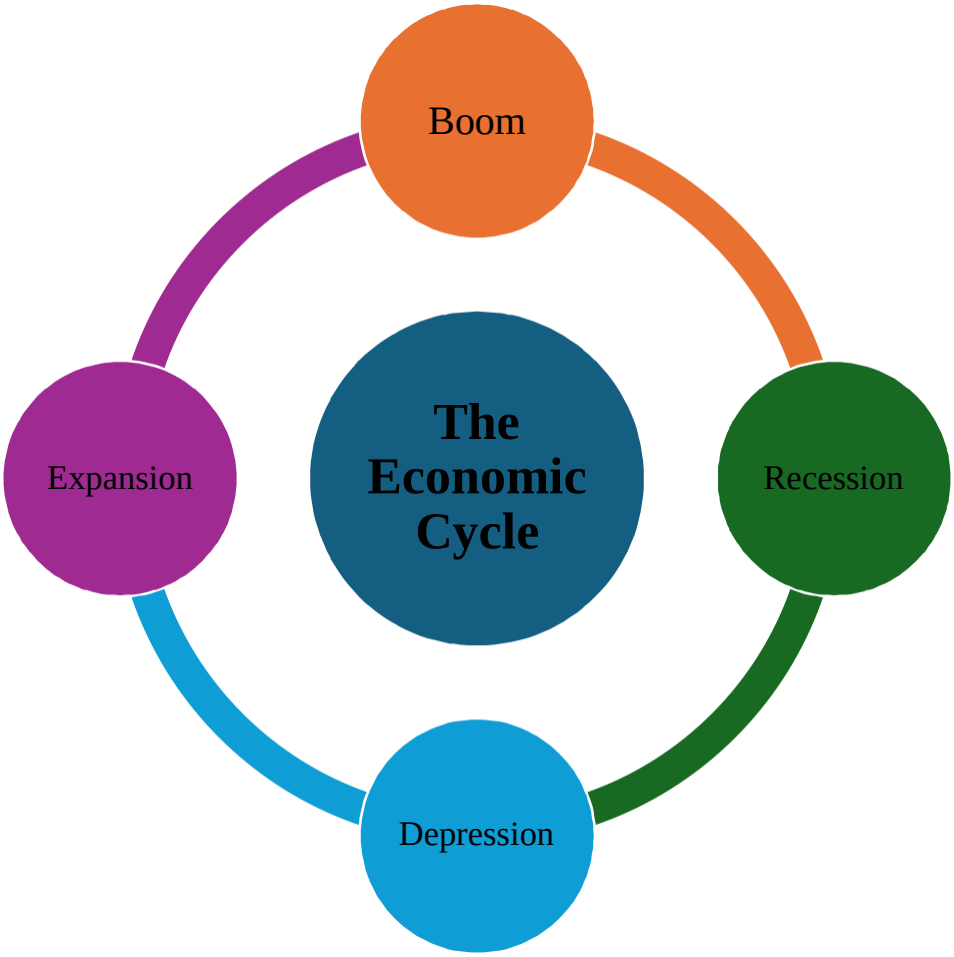

Figure 3.1 The Economic Cycle

The bank lending activities are often linked with the economic cycle. Many researchers find that bank lending activities increase during the economic expansion and boom phases. It is attributed to the increased demand for bank finance to facilitate an increased rate of economic activities. However, this procyclicality of bank loans during the expansionary period poses a threat during the recession period (Caporale et al., 2014b). The study based on the Italian banking sector studies whether the more than proportionate increase in bad loans in the recessionary period can be explained by the excessive lending growth during the boom period in the economy. The study indicates that the effect is significant in the case of the advances made to the firm. The recessionary pressure has the most impact on the businesses in the

economy which are worst affected and struggle to survive and hence default on loan repayments.(Leroy & Lucotte, 2019). A recent study also studies the European banking sector for the impact of credit procyclicality. They study that banks become lax for loan loss provisions during the boom period. They say that the loan loss provisions are negatively correlated with GDP growth. Lack of proper provisioning creates a sense of increased profitability in the short run, but during the recession period, when the NPLs exert pressure, lack of provisioning puts immense pressure on the bank's balance sheets.

The cyclicity of lending activities is well-established in the banking literature. This pattern of lending is linked to macroeconomic conditions. A similar pattern is also seen in the NPLs (Goel, 2018; Kanazir, 2023). The cyclicity of the behaviour of these variables, apart from being linked to the macroeconomic conditions, is also linked to each other. Excessive lending in the expansionary period leads to an overextended economic credit supply. With an excess of funds in hands, it sometimes becomes a case that the firms cannot manage the funds properly or there is a diversion of funds, i.e., funds are not put to use for the sanctioned purpose, which leads to default in repayments leading to an increase in NPLs.Figure 2.3 shows the pattern of banking activities during different phases of the economic cycle.

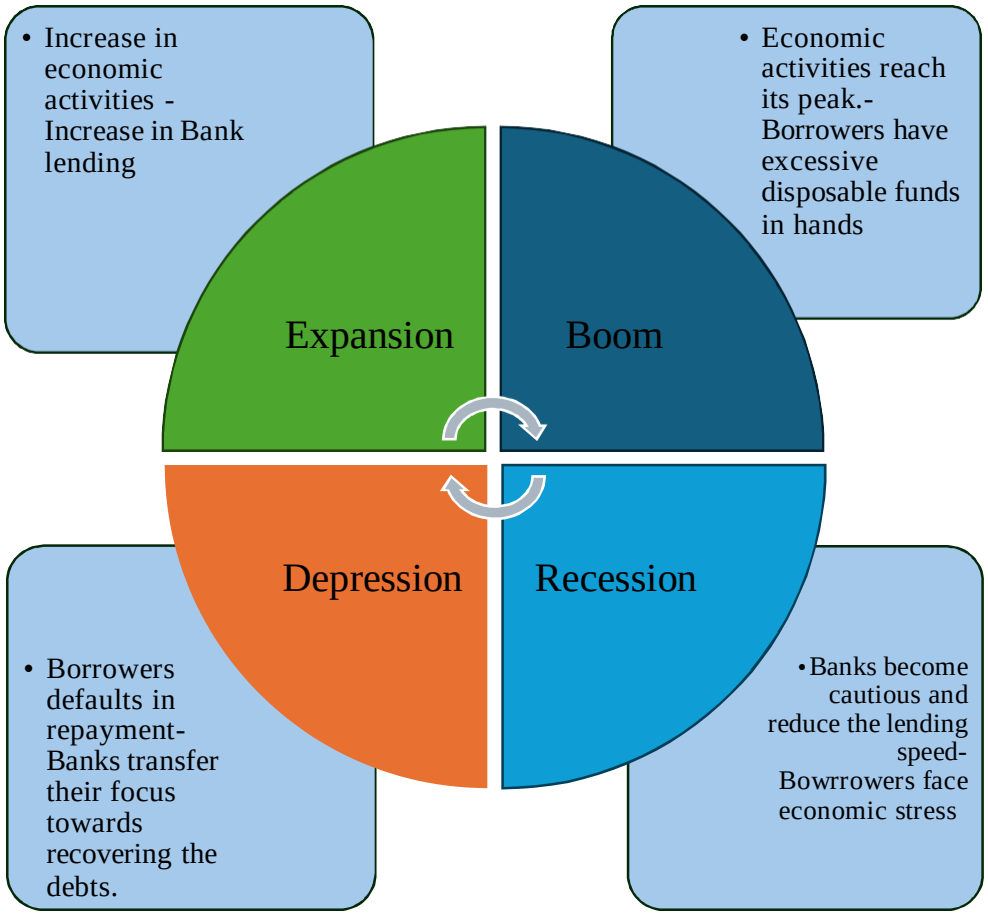

Figure 3.2 Linking the bank's business cycle with the economic cycle.

3.1.3 Theoretical framework development

The theoretical framework gives a basis for conducting an empirical study. After discussing the four hypotheses stated in section 3.1.1 and the procyclicality of loans and NPLs in section 3.1.2, we now try to connect the dots to create a framework and subsequently state our hypotheses.

- **Loan growth and Bank Profitability**

Loan growth increases bank profitability by increasing the interest income. However, an excessive increase in loan growth leads to a negative impact on profitability. We hypothesize

that “bad management”, “Skimping” and “moral hazard” come into play during the phases of an economic boom. Banks try to seize the opportunity and increase lending activities by ignoring the pre-sanction and post-sanction parts due to increased NPLs. (Adelopo et al., 2018a). Also, examine the factors that influence banks' profitability, paying particular attention to the periods before, during, and post the financial crisis, and conclude that the loan growth negatively impacts the profitability during the crisis and after the crisis period, thus examining how the economic cycle affects the factors that determine profitability. Our study is based on the Indian banking sector and the Indian economy is an emerging economy that needs financial assistance for budding businesses to grow, which is beneficial for the economy as a whole as well as for the Bank's profit, but at the same time, we can see the effects of bad lending decisions on the bank’s balance sheet from the year 2015 onwards which have been burdened with the mounting load of NPAs. (Le, 2020b) examines a quadratic relationship between loan growth and profitability in the Vietnamese banking sector. The study suggests a U-shape relationship between profitability and loan growth and suggests that regulators be vigilant in times of massive loan growth. Our hypothesis in this study is that loan growth has a non-linear relationship with profitability. While a steady growth of loans positively impacts profitability, excessive lending may negatively impact the bank’s profit and thus have an inverse U-shaped relationship.

H_1: Loan growth and profitability have an Inverse U-shaped relationship.

- **Loan Growth and Non-performing Loans**

Periods of excessive lending are generally periods of boom in the economy. During which the general financial well-being is good in the country, and thus the NPLs are on the lower side. However, once the excessive lending starts, the “bad management”, “Skimping,” and “moral hazard” play their part, and the quality of loans deteriorates, causing NPLs to increase.

(Bhowmik & Sarker, 2021) their study based on SAARC nations finds that loan growth induces bank risk. This bank risk worsens when the speed of lending increases. Hence, Assuming a linear negative relationship between NPLs and loan growth does not fully justify the nature of the relationship between these two variables. Our study assumes a non-linear U-shaped relationship between loan growth and NPLs.

H_2: Loan growth and NPLs have a U-shaped relationship.

- **Non Performing Loans and Profitability**

Different researchers have different findings and views regarding the relationship between NPLs and profitability, as discussed in section 2.2.6. NPLs impact profitability by increasing the amount of provisioning to be done. Also, the capital requirements of the banks increase as the assets become riskier, which entails a considerable capital cost to the banks, thus reducing profitability**.** We assume a negative relationship between NPLs and profitability.

H_3: NPLs have a negative impact on profitability.

- **Mediating role of NPLs**

As stated in Hypothesis 1, the Loan growth has a non-linear relationship with the profitability. The loan growth first increases the profits by increasing the interest income, but the negative manifestations of the loan growth come into play once the excessive lending starts. These negative manifestations generally link to the NPLs.The extra costs involved in recovering loans and monitoring the deteriorating loans mediate the relationship between loan growth and profitability.

H_4: NPL mediate the relationship between the Loan growth and profitability

Figure 3.3 shows a pictorial representation of our hypotheses in the form of a theoretical framework. We also include some control variables in our study to segregate the impact of our three variables of the study on each other. We discuss each of these variables in detail in the next section.

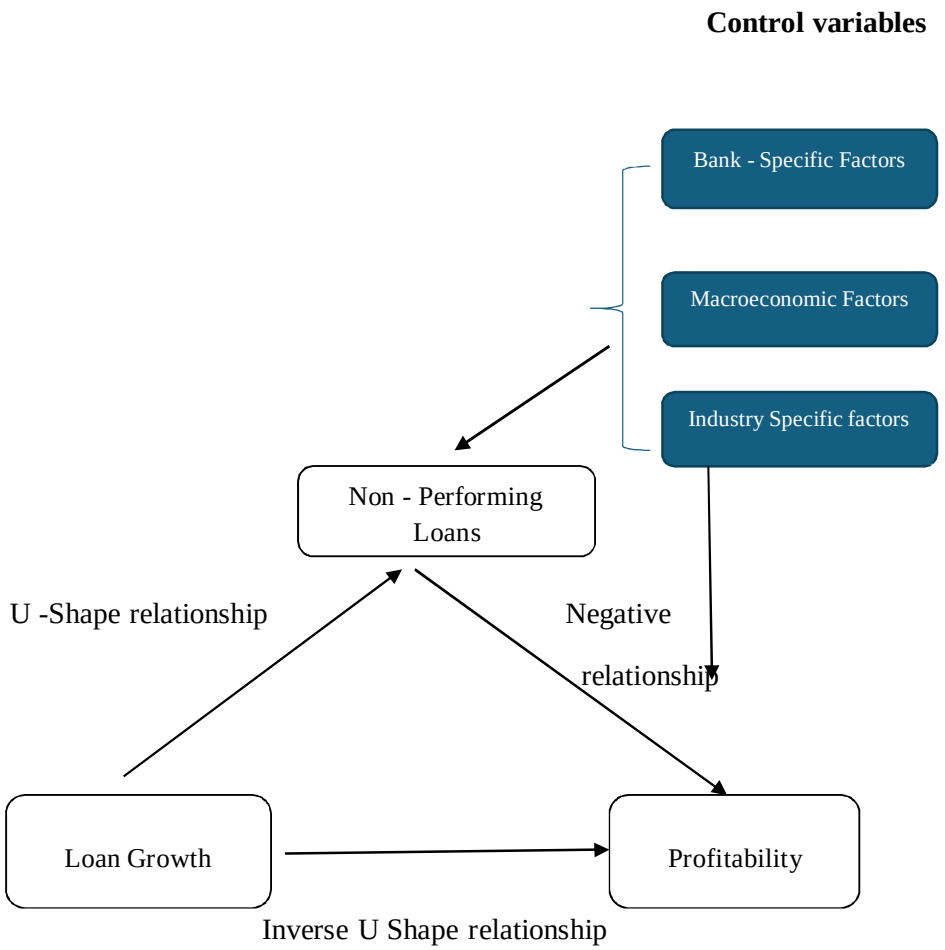

Figure 3.3 Theoretical framework for the relationship between - Loan growth, NPLs, and profitability.

3.2 Data and Variables

3.2.1 Data

This paper analyses the financials of 12 public and 17 private sectors from 2006 to 2021.The period selected for the study includes pre crisis , crisis and post crisis period hence is relevant for the study..

Therelevant data has been extracted from the RBI's website; hence, all the data is reliable and as

per all the disclosure norms. Some banks have been dropped from the study scope due to the unavailability of financials for the said period. Table 3.1 shows a detailed list of our sample.

Public sector banks	**Private sector banks**
Bank of Baroda	Axis Bank
Bank of India	City Union Bank Limited
Bank of Maharashtra	CSB Bank Limited
Canara Bank	DCB Bank Limited
Central Bank of India	Federal Bank
Indian Bank	HDFC Bank Ltd.
Indian Overseas Bank	ICICI Bank
Punjab and Sind Bank	IDBI Bank Limited
Punjab National Bank	IndusInd Bank
State Bank of India	Jammu & Kashmir Bank Ltd
UCO Bank	Karnataka Bank Ltd
Union Bank of India	Kotak Mahindra Bank Ltd
	RBL Bank Limited
	South Indian Bank Ltd
	Tamilnad Mercantile Bank Ltd
	The Dhana Lakshmi Bank Ltd
	Yes Bank Ltd.

Table 3.1 List of banks included in our sample segregated based on ownership

Notes to Table 3.1 - Bandhan Bank and IDFC First Bank have been excluded from the study as both were incorporated in 2014. Also, the study has not included Karur Vysya Bank and Nainital Bank due to some financial ratios/data unavailability.

During our study, many mergers took place in the Indian banking sector. Table 3.2 shows the details of all such mergers. The banks' financials have been merged per extant guidelines for this research.

S.no	Parent Bank	Merged Bank	Merged Bank	Year
1	Bank of Baroda	Dena Bank	Vijaya bank	2019
2	ICICI	Bank of Rajasthan		2010
3	Kotak Mahindra	ING Vysya		2016
4	Punjab National Bank	Oriental bank	United Bank	2020
5	Canara Bank	Syndicate Bank		2020
6	Indian Bank	Allahabad Bank		2020
8	Union Bank of India	Andhra Bank	Corporation Bank	2020
9	State Bank Of India	State Bank of Saurashtra		2010
		State Bank of Indore		2012
		State Bank of Bikaner and Jaipur		2018
		State Bank of Hyderabad		2018
		State Bank of Mysore		2018
		State Bank of Patiala		2018
		State Bank of Travancore		2018
		Bhartiya Mahila Bank		2018

Table 3.2 Details of Bank Mergers in the Indian Banking Sector during the period of our study

3.2.2 Variable Description

We use a set of key and control variables to study the three aspects of banking – loan growth, NPLs, and profitability. Table 3.4 briefly describes these variables, followed by a detailed discussion.

Variable	Symbols	Description	Previous studies
Key Variables			
Loan Growth	∂GA	Percentage Change in Gross Advances	(Abedifar et al., 2018a; Ibrahim & Rizvi, 2018)
NNPA to GA	NNPA to GA	Net NPA to Gross Advances	(Abedifar et al., 2018a; Ibrahim & Rizvi, 2018)
ROA	ROA	Return on Assets	(Almaqtari et al., 2019b; Vo, 2018a)
Bank-Specific Control Variables			
Asset Size	Ln TA	Log of Total Assets	(Foos et al., 2010c; Mostak Ahamed, 2017c; Sufian & Habibullah, 2009c)
Loan Portfolio/Liquidity Risk	LR	Gross Loans to Assets	(Almaqtari et al., 2019b).
Capital	CAP	Capital	(Abedifar et al., 2018)(Dang, 2022)
Macroeconomic Control Variables			
GDP growth	GDP	Growth in Gross Domestic Product	(Almaqtari et al., 2019b; Sufian & Habibullah, 2009c; Vo, 2018b)
Inflation	INF	Real Inflation rate	(Almaqtari et al., 2019b; Sufian & Habibullah, 2009c; Vo, 2018b)
Industry-Specific Control Variables			
Industry NPA	INPA	Average NPA of all banks	(Vo, 2018b)
Industry profit	IP	Average ROA of all banks	(Vo, 2018b)

Table 3.3 - Variable description

KEY VARIABLES

- **Loan Growth - YoY growth of loans -** Growth in the loan outstanding of a bank on a YoY basis shows a trend and gives an overview of the bank's lending behaviour. It is calculated as the Percentage Change in Gross Advances denoted by ∂ GA. For capturing the banks' excessive lending, we use the quadratic terms of the loan growth, i.e. ∂GA^2.

- ***Non-Performing Loans* - Ratio of NNPA to GA -** The Non-performing Asset (NPA) figure that appears in the balance sheet is net NPA, i.e., NPA at the beginning of the period and additions during the period minus recoveries made in NPA accounts minus the provisions. The Net NPA to Gross Advances ratio is the ratio of NPLs. Recoveries made by the bank during the period affect the NPA level. To understand the impact of NPLs, we must consider the following ratio.

  NNPAto GA= Net Non-Performing Assets / Gross Advances

- ***Profitability* - Return on assets (ROA):** A bank's profitability is measured by different researchers through different ratios. Different profitability ratios depict the efficiency of an organization through different metrics. The most widely used metric for profitability is the return on assets, which has been used as a proxy for profitability in this study. Return on Assets is a profitability metric that helps measure profitability concerning its total assets.

$$\text{Return on Assets} = \frac{\text{Net income}}{\text{Total assets}}$$

CONTROL VARIABLES

A control variable is a variable that is not the variable of interest in the study, but it affects the dependent variable; hence, it is necessary to study its effect to understand the real impact of the independent variable. We have used the following variables as control variables:

BANK SPECIFIC CONTROL VARIABLES

- **Loans Ratio/Liquidity Risk:** This ratio tells us whether the bank is more inclined towards lending, i.e., whether the significant portion of the bank's assets is comprised of the amount advanced to customers. This ratio is also a proxy of size in terms of the advanced portfolio of the bank. The ratio has been calculated as the Gross loans-to assets ratio.

- **Total assets:** It has been used as a control variable controlling the effect of varied sizes of the different banks. Bigger banks may be more profitable by size; hence, we must account for this effect to understand the real effect of loan growth and asset quality on profitability. It has been proxied by log(total asset).
- **Capital:** Capital has also been chosen as a control variable. Highly capitalized banks have larger stakes in the profits than the smaller ones. Hence, Capital may play a role in this framework.

MACROECONOMIC CONTROL VARIABLES

- **GDP Growth:** The GDP Growth Rate expresses how much the nation's GDP has changed from one time to another. The amount of change is expressed as a percentage (%), used to assess the nation's economic health and potential for future growth. The GDP growth rate has been used as a proxy for a country's macroeconomic environment, which might impact our key variables.
- **Inflation:** Inflation is the rate at which prices rise over a specific period. Inflation is defined as a general increase in prices of commodities necessary for sustaining life in a country. Inflation has also been used as a proxy for the macroeconomic environment.

***GDP growth and Inflation rates are sourced from the World Bank's website.**

INDUSTRY-SPECIFIC CONTROL VARIABLES

The Banking Industry scenario prevailing in the country at a particular point in time is also a factor that can impact our key variables.

- **Industry Profit:** The banks' average profit in a particular year is calculated. For public sector banks, we take the average profits of all the public sector banks and, similarly, the private sector banks.

- **Industry NPA:** Industry NPA is calculated as the average NPA for all the banks in a particular year. For public sector banks, we take the average NPA of all the public sector banks and, similarly, the private sector banks.

3.3 Research Methodology

The paper analyses the panel data of India's public sector and private sector banks. Our key variables –Loan Growth, ROA, and NNPAtoGA have a risk of having an inherent endogeneity and simultaneity. To resolve this issue, we use the System Generalised Method of Moments. This section discusses all the methods and tools used in this study.

3.3.1 System Generalised Method of Moments (SYS-GMM)

Endogeneity bias poses a risk of rendering the results of regression obsolete. When used with endogenous variables, a random effect or ordinary least square regression may provide coefficients with a wrong sign and deliver wrong results. (Ullah et al., 2018) in their study try to showcase the endogeneity bias in panel data study and resolve it using the Generalised Method of Moments (GMM). The results of a panel data study using the OLS and random effects regression are compared with the GMM regression, and significant differences were found. We use the GMM suggested by (Arellano & Bover, 1995; Blundell & Bond, 1998), which solves the endogeneity and simultaneity issue in the variables loan growth, NPLs, profitability, and other bank-level indicators. The GMM can solve data issues like endogeneity, reverse causality, and simultaneity bias. This technique is specially designed for dynamic panel data analysis where the effect of a variable in one period has an inherent effect from the previous period. The dynamic panel data model in GMM incorporates the lags of dependent variables as explanatory factors used as instruments to control the endogenous relationship. The GMM has two types – The first difference or one-step GMM, and the

Second difference or two-step GMM, also called SYS-GMM. The use of SYS-GMM is preferred over the First difference GMM for avoiding the problem of potential data loss. The one-step GMM deducts the past value of the variable from the current value, reducing the number of observations. The SYS-GMM uses forward orthogonal deviations, i.e., subtracting the mean value of all subsequent observations of a particular variable from its present value. SYS-GMM is preferred for balanced panel data as it provides more efficient variable estimates (Arellano & Bover, 1995). Based on the suggestions by the abovementioned study, we form the following set of regression equations for our model using the SYS-GMM :

$$ROA_{it} = \beta_0 ROA_{i.t-1} + \beta_1 NNPAtoGA_{it} + \beta_2 \delta GA_{it} + \beta_3 \delta GA^2{}_{it} + \beta_4 C_{it} + \beta_5 IP_{it} + \beta_6 M_{it} + s_{it} \quad (1)$$

$$NNPAtoGA_{it} = \beta_0 NNPAtoGA_{i,t-1} + \beta_1 ROA_{it} + \beta_2 \delta GA_{it} + \beta_3 \delta GA^2{}_{it} + \beta_4 C_{it} + \beta_5 INPA_{it} + \beta_6 M_{it} + s_{it} \quad (2)$$

Here, regression factors of explanatory variables are represented by β_0, β_1, β_2, β_3, β_4, β_5, and β_6 and s_{it} represents the unobservable properties. The dependent variable is NNPAtoGa as a proxy for non-performing loans in equation 1 and ROA as a proxy for profitability. $NNPAtoGA_{i,t-1}$ *and* $ROA_{i.t-1}$ is one period lag of the dependent variable δGA *and* δGA^2 is the quadratic term used to check for the non-monotonic relationship between Loan growth on the NPLs and bank profitability? C_{it} is the Vector of the bank-specific characteristics used as control variables, and M_{it} is a vector of macroeconomic variables described in Table 3.4.

3.3.2 Arellano Bond test for autocorrelation

The SYS-GMM does not require any specifications for data to be normal or stationary; such diagnostics tests need to be run before starting with the SYS-GMM estimation after the

estimation, however, we check for autocorrelation. It is similar to checking the model fit. Auto-correlation, also known as serial correlation, occurs when the regression residuals are correlated with each other. In other words, it occurs when the errors in the regression are not independent of each other. We run the Arellano-Bond test on first and second differenced errors for autocorrelation. The null hypothesis for the Arellano Bond test for autocorrelation is "There is no autocorrelation in the disturbances". The Arellano-Bond test for AR(1) and the Arellano-Bond test for AR(2) is done. The first-order serial correlation is expected due to the lagged dependent term and is not a problem. The model is valid if the absence of the second- order serial correlation in disturbances (the null hypothesis) is not rejected. (Arellano & Bover, 1995).

3.3.3 Hansen Sargan test for overidentification of variables

Overidentification means that the number of endogenous regressors is less than the number of instruments in the equation. If an equation or model is overidentified, tests for overidentification restrictions must be applied to check the validity of the instruments. (Sargan, 1958) The Hansen-Sargan test for over-identifying limitations assesses the consistency of our estimator. The null hypothesis for the Hansen-Sargan test is "All instruments are valid". The insignificant Sargan statistics for over-identification confirm the suitability of our model.

3.3.4 Lind Melhum U Test

By incorporating the quadratic terms of the loan growth proxies, we investigate the possibility of a non-monotonic connection. We first check the regression coefficients to establish a U-shaped relationship between the variables. If the sign of the original and the quadratic terms are opposite, it implies a U-shaped relationship. Then, to confirm the significance of the

relationship and the extreme points U test proposed by (Lind & Mehlum, 2010) is performed using the three-step procedure proposed by (Lind & Mehlum, 2010) and recommended by (Haans et al., 2016) for social science studies by estimating the following model:

$$NNPAtoGA = \alpha\ \partial\mathrm{GA} + \beta\,\partial\mathrm{GA}^2 + z_{\mathrm{i}} + \mathrm{s}_{\mathrm{i}t}$$

Next, we test the joint hypotheses: -

$$H_0 \quad (\alpha + \beta^2\ \partial\mathrm{GA}_{min} \leq 0\ )\ \cup\ (\ \alpha + b^2\,\partial\mathrm{GA}_{max}\ \geq 0\ )$$

Against the alternate hypotheses

$$H_1 \quad (\alpha + b^2\ \partial\mathrm{GA}_{min} > 0\ )\ \cup\ (\ \alpha + b^2\,\partial\mathrm{GA}_{max} < 0\ )$$

the $\partial\mathrm{GA}_{min}$ and $\partial\mathrm{GA}_{max}$ represent the minimum and maximum values of Loan Growth, respectively. The rejection of Null Hypotheses will confirm a U-shaped relationship between the variables. Similarly, we test the non-monotonic U-shape relationship between Profitability and Loan Growth by estimating the following model:

$$ROA = \alpha\ \partial\mathrm{GA} + \beta\,\partial\mathrm{GA}^2 + z_{\mathrm{i}} + \mathrm{s}_{\mathrm{i}t}$$

Next, we test the joint hypotheses: -

$$H_0 \quad (\alpha + \beta^2\ \partial\mathrm{GA}_{min} \leq 0\ )\ \cup\ (\ \alpha + b^2\,\partial\mathrm{GA}_{max}\ \geq 0\ )$$

Against the alternate hypotheses

$$H_1 \quad (\alpha + b^2\ \partial\mathrm{GA}_{min} >\ 0\ )\ \cup\ (\ \alpha + b^2\,\partial\mathrm{GA}_{max} < 0\ )$$

So, the rejection of null hypotheses will confirm that there is a U-shaped relationship between the variables.

3.3.5 Quantile regression

Quantile regression is used (Koenker & Bassett, 1978) as it offers an alternative, widely used way to assess possible heterogeneity in a dataset. Compared to other regression methods,

which usually assume that relationships between independent and dependent variables as constant across all levels, quantile regression offers an alternative. Quantile regression is not a regression estimated on quantile or subsample data. The objective of regression is to reduce the differences between the values that the regression line predicts and the values that are observed. On the other hand, quantile regression attempts to minimize the weighted distances by variably weighing the differences between the predicted and actual values by the regression line. (Cook & Manning, 2013).

3.3.6 Mediation Analysis – Sobel test.

Mediation analysis tests the impact of an independent variable on a dependent variable in the presence of a third variable known as a mediator. The mediation is said to be present when it influences the relationship between the Independent Variable (IV) and Dependent Variable (DV). In simple terms, the mediation exists when the following conditions are fulfilled :

- The IV has a significant impact on the mediator.
- The IV also has a significant direct impact on the DV
- The mediator significantly impacts the DV
- The introduction of the mediator variable influences the relationship between IV and DV.

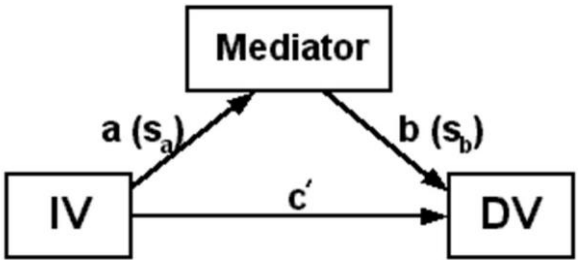

Figure 3.4 Mediation illustration

Figure 3.4 shows a path diagram of a model that shows mediation. The path coefficients of the three relationships are a, *b*, and *c'*. s_a and s_b are the standard errors of these coefficients calculated through regression analysis.

To conduct the Sobel test (Sobel, 1982) we use the following based on the Z score.

Z value for Sobel Test = $a*b/\sqrt{(b^2 * s_a^2 + a^2 * s_b^2)}$

When the Z value is significant for the relationship between the IV and Mediator, DV and Mediator, and IV and DV, It is said that the mediation exists, but it is partial. If the Z value is significant for only IV and mediator and DV and Mediator, the mediation is said to be complete. If either IV and mediator or DV and mediator are not significant, then it is said that mediation doesn't exist.

After the mediation is confirmed, we test the degree of mediation effect by using the following formula:

The ratio of the indirect effect to the total effect (RIT) = (Indirect effect / Total effect)

The ratio of the indirect effect to the direct effect (RID) = (Indirect effect / Direct effect)

3.4 Research tools

Stata 16.1 has been used for complete analysis, including SYS GMM, U test, and mediation analysis.

Chapter 4
Empirical Analysis, Results & Discussions

4.1. Descriptive analysis

This chapter presents and discusses the results of our study. Before reporting the results we first have a look at our variables. The following subsections report the summary statistics, correlation analysis, and the stationarity test of the variables.

4.1.1 Summary statistics

Table 4.1 shows the summary statistics of our variables. The table shows the mean and standard deviation for each variable in all three groups. – All banks, Public sector banks, and private sector banks. The All Banks group has 29 panels and 464 observations. Public sector banks have 12 panels and 192 observations; private sector banks have 17 panels and 272 observations. The mean value of NNPA to GA in Public sector banks is much higher than that of private sector banks, with a higher standard deviation. The ROA of public sector banks is on the lower side; similar is the case with loan growth.

	All Banks		Public Sector		Private Sector	
Variable	**Mean**	**Std. Dev.**	**Mean**	**Std. Dev.**	**Mean**	**Std. Dev.**
NNPA to GA	2.38	2.32	3.53	3.09	1.57	1.57
ROA	0.61	0.84	0.24	0.88	0.87	0.82
∂GA	0.18	0.18	0.14	0.14	0.21	0.20
Ln TA	11.68	0.73	12.70	0.55	10.95	0.83
LR	0.59	0.05	0.59	0.05	0.59	0.05
CAP	1147.12	1585.78	2056.79	2240.35	505.01	871.09
GDP	0.06	0.04	0.06	0.04	0.06	0.04
INF	0.07	0.03	0.07	0.03	0.07	0.03
IP	1.10	0.47	1.10	0.47	1.06	0.47
INPA	1.84	0.91	1.74	0.91	1.62	0.91

Table 4.1 Summary statistics of the variables – Mean and Standard Deviation

4.1.2 Correlation Analysis

The correlation analysis shows how a variable is related to another variable. A high degree of correlation in the variables can hamper the results of the empirical analysis. Table 4.2 shows the correlation table of the All Banks panel. A correlation coefficient of 0.8 or more as an absolute value is considered a high correlation. None of the values in our sample is more than 0.80 except the correlation between IP and INPA. These variables have not been used together; hence do not pose any problem for our results.

	NNPA to GA	ROA	∂GA	Ln TA	LR	CAP	GDP	INF	IP	INPA
NNPA to GA	1.00									
ROA	-0.75	1.00								
∂GA	-0.53	0.51	1.00							
Ln TA	0.29	-0.15	-0.29	1.00						
LR	-0.25	0.31	0.08	0.15	1.00					
CAP	0.42	-0.50	-0.26	0.39	-0.36	1.00				
GDP	-0.08	0.22	0.14	-0.11	-0.03	-0.25	1.00			
INF	-0.41	0.31	0.29	-0.19	0.04	-0.23	0.41	1.00		
IP	-0.52	0.44	0.37	-0.29	-0.04	-0.31	0.36	0.68	1.00	
INPA	0.59	-0.38	-0.38	0.28	0.07	0.17	-0.06	-0.65	-0.86	1.00

Table 4.2 Correlation Matrix

4.1.3 Unit root test

The Unit root test is the test of stationarity of the variables. We use Stata to conduct Levin Lin Chu and the Harris Tzavallis test for stationarity. The Null hypothesis for both tests is that the panel consists of unit roots. Table 4.3 shows the result for all the variables of our model using both tests. Some of the variables in our study contain unit roots and are not stationary – LnTA, LR, CAP, and INPA.As we have chosen SYS-GMM for the dynamic panel data analysis, the non-stationarity of the variables does not affect our results.

	Levin Lin chu		Harris-Tzavalis	
	t stat	**P value**	**t stat**	**P value**
NNPA to GA	-3.1764	0.0007	0.9353	0.000
ROA	-2.5434	0.0055	0.805	0.000
∂GA	-5.029	0.0000	0.7488	0.000
Ln TA	-10.8277	0.0000	1.0116	0.752
LR	-1.509	0.0656	1.0012	0.528
CAP	-0.0959	0.4618	0.9912	1.000
GDP	-6.4435	0.0000	0.7131	0.000
INF	-3.654	0.0001	0.9191	0.000
IP	-1.5921	0.0557	0.9515	0.002
INPA	-4.5048	0.0000	0.9797	0.115

Table 4.3 Unit Root Test

4.2 Results & Discussion

This section presents the results of the SYS-GMM, Mediation analysis, and Quantile regression.

4.2.1 SYS-GMM results

As discussed in section 3.3.1, we use 2 regression equations to run the GMM model for our dynamic panel data to prove our first three hypotheses.

Table 4.4 below shows the results for equation 1 – which checks the relationship between loan growth and profitability and NPLs and profitability, and Table 4.5 shows the results for equation 2, which establishes the relationship between loan growth and NPLs.

Table 4.4 has ROA as the dependent variable. The relationship of the DV with the explanatory variables is explained in this section. The macroeconomic variables - GDP growth as an indicator of the contribution to the ROA of a bank are consistent with the existing literature (Sufian & Habibullah, 2009c) (Gupta & Mahakud, 2020) GDP growth has a positive impact on the profitability of the banks but is contradicted by (Al-Homaidi et al.,

2018). This suggests that in periods of a boom in the economy, the banking sector also flourishes with the boom in other sectors of the economy. Inflation, on the other hand, has a negative impact on the bank's profitability, which may be attributed to the decreased liquidity in the hands of the public, thus reducing the savings and investment level in the economy, indirectly impacting the bank's profitability. The results are also supported by the results in the study (Al-Homaidi et al., 2018; Faniband, 2020; Madugu et al., 2020).

Amongst the bank-specific control variables, the bank size and capital significantly impact profitability. The bank size, which is proxied by the log of total assets, has a positive impact on the profitability of the bank, which is supported by existing literature (Al-Homaidi et al., 2018; Almaqtari et al., 2019b). The larger size of the asset side of the bank's balance sheet it, earns more profits.

As we have already seen, rising GDP has a positive impact on profitability; rising economic activities on account of rising loan growth will lead to improved profitability, but the rise in lending beyond a threshold where the funds cannot be put to proper use and thus diversion of funds eventually turns the account NPA which negatively impacts the profitability. Our GMM estimation results also support our hypotheses in the case of data of all banks and private sector banks where the non-quadratic terms of loan growth have a +ve sign and the quadratic term has a -ve sign indicating an inverse U shape relationship, which is again tested by U shaped test suggested by (Lind & Mehlum, 2010) with an extreme point and lower and upper bound interval as shown in Table VI. As in the case of public sector banks, the relationship is non-monotonic but with reverse signs, thus creating a U-shaped relationship. We verify this relationship further using the quantile regression in the subsequent sections.

Table 4.4 also explains the relationship between the NPLs and profitability. NPLs negatively impact the profitability of the whole banking sector and the public and private sector banks individually. This relationship is supported by the studies of (Ozili & Arun, 2022)(Bansal et al., 2018) (Gaur & Mohapatra, 2021a; Laryea et al., 2016a). Based on the literature review and empirical results, our study emphasises the existence of a negative relationship between these two variables. Any increase in the NPLs increases the provisioning requirement, a direct charge-off of the bank's profits.

The SYS-GMM uses the lags of the instrumental variables for estimating the model. Different lags are used for the three subgroups' NNPA to GA and Loan growth. For NNPA to GA All bank's sample uses 0 lag, Public sector banks use 0 lag, and private sector bank uses 1 lag. For Loan growth, the All banks sample uses 4 lags, Public sector bank uses 1 lag, and private sector banks use 2 lags. The different lags imply how much time, in terms of years, it takes for the impact to take place.

Level of Sig. - ***p<0.01, **p<0.05, *p<0.1

Table 4.4 – Regression results for effects of NPLs and Loan growth on Profitability using GMM estimation.

	All Banks		**Public Sector Bank**		**Private sector Bank**	
DV-ROA	**Model 1**	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**
NNPA to GA	-0.1628***	-0.1293***	-0.1155***	-0.1245***	-0.2165***	-0.2322***
∂GA	0.4991***	1.3811***	0.0631	-0.8631*	0.7715***	1.6748***
∂GA^2		-0.3670***		2.7983**		-0.5369***
IP	0.2672***	0.3488***	0.8258***	0.7925***	0.2729***	0.1966***
Ln TA	0.0668	0.1719***	0.0353***	0.0455*	0.1902***	0.1644***
LR	1.1073	1.8392***	0.4201	0.936	1.9949	1.4533
CAP	-0.0001***	-0.1864***	-0.0001***	-0.0001***	-0.0003***	-0.0003***
GDP	1.4692	2.2052***	1.6050*	1.7735**	1.6382	3.1851***
INF	-1.9196	-2.5448***	-7.902***	-7.3768***	-0.9733	-1.8911
_cons	-0.5426	-1.5477***	-0.285	-0.6883	-2.3244***	-1.7661***
Hansen Sargan Test	0.072	0.361	0.94	0.993	0.246	0.255
Arellano Bond test 1st diff	0.016	0.009	0.033	0.035	0.118	0.097
Arellano bond test 2nd diff	0.12	0.128	0.257	0.23	0.055	0.064
No. of Observations	464	464	192	192	272	272
U shape test		Inverse U shape		U shape		Inverse U shape
Extreme point		1.882		0.1542		1.5596
Lower Bound		-0.2901		-0.2256		-0.2901
Upper bound		2.1631		0.5628		2.1631

Table 4.5 discusses the results of Equation 2, which explains the impact of loan growth on NPLs. The Macroeconomic variables GDP and inflation significantly impact PLs(Arrawatia et al., 2019). As per our results, the increase in GDP increases the NPLs, which is consistent with the (Hajja, 2022) but contradicts (A. Ghosh, 2015; Umar & Sun, 2018). NPLs' relationship with inflation is negative, i.e. with an increase in inflation, the NPLs decrease. The result suggests that the NPLs decrease significantly as the money supply increases because borrowers tend to get funds at a relatively lower cost, which boosts their capacity to repay their debts. This theory is backed by the empirical results of the studies (Abbate A & Thaler, 2019; Anita et al., 2022; Hajja, 2022)

Apart from capital, size, and liquidity, the bank-specific control variables have mixed results. Capital has a negative impact on NPL, i.e., well-capitalized banks tend to have lower NPLs.Existing literature supports this result(Abedifar et al., 2018; Choudhary & Limodio, 2022). Next is the impact of profitability on NPLs; our regression results prove that profitability has a negative impact on NPLs, which is supported by existing literature (A. Ghosh, 2015; Kumar et al., 2018).

The regression coefficients of the explanatory variable loan growth proxied by δGA and δGA^2 are significant and are as per our hypotheses. Model 8, model 10, and model 12 show the regression result for the quadratic terms of loan growth for all bank samples, the public sector banks sample, and the private sector bank sample. The signs of the coefficients of variable δGA and δGA^2 change from -ve to +ve, indicating that loan growth, in the beginning, diminishes the NPLs of the banks, but excessive growth can lead to an increase in NPLs.The Lags used for the Loan growth and NNPA for All Banks sample is 2, for public-sector banks is 1, and for private-sector banks is 3. It can be comprehended that the negative

impact of loan growth takes longer to manifest in private-sector banks than in public-sector banks.

	All Banks		Public Sector Banks		Private Sector Banks	
DV- NNPA to GA	**Model 7**	**Model 8**	**Model 9**	**Model 10**	**Model 11**	**Model 12**
ROA	-1.4315***	-1.2748***	-1.1970***	-0.2434	-1.2093***	-1.1458***
∂GA	-1.2341**	-2.7784***	-7.6326***	-33.3238***	-0.3079	-0.7283***
∂GA^2		1.2294***		89.9969***		0.3632***
INPA	0.8480***	0.77690***	1.1704***	1.6195***	0.3642***	0.4064***
Ln TA	0.2308***	0.1949***	-0.3217***	0.2544	0.1216**	0.1034**
LR	-6.0521***	-4.2539***	1.0372	17.6729***	-2.2243	-1.6897
CAP	-0.0001	-0.0004	-0.0001**	0.0003***	-0.0002***	-0.0001***
GDP	5.0464***	4.4844***	5.9407**	5.9232**	3.3508***	3.3139***
INF	-5.8512**	-4.5489*	-11.9651***	1.5187	-3.2291**	-0.7245
_cons	3.0500***	2.4483***	6.7636***	-13.0493**	2.1010***	1.7348***
Hansen Sargan Test	0.308	0.192	0.2345	0.945	0.470	0.371
Arellano Bond test 1st diff	0.018	0.02	0.008	0.002	0.517	0.592
Arellano bond test 2nd diff	0.062	0.064	0.085	0.917	0.191	0.199
No. of Observations	464	464	192	192	272	272
U shape test	**U shape**		**U shape**		**U shape**	
Extreme point	1.13		0.1851		1.002	
Lower Bound	-0.2901		-0.2256		-0.2901	
Upper bound	2.1631		0.5627		2.1631	

Level of Sig. - ***p<0.01, **p<0.05, *p<0.1

Table 4.5 – Regression results for effects of Loan growth on NPLs using GMM

Model Validation

To ensure the model and the instrument's validity in SYS- GMM, we conduct the Arellano Bond test for autocorrelation and the Hansen Sargan Test for the overidentifying instruments. The results for both these tests are also provided in Table 4.5.

The Arellano Bond Test for serial correlation in 2nd order residuals is insignificant; thus, we accept the null hypothesis, i.e. no serial correlation exists.

Also, the Hansen- sargan test results also accept the null hypothesis in all the samples. Thus proving that all the instruments are valid.

Lind Mehlum U test

The quadratic terms of loan growth have been used in the model to see whether the non-linear relationship hypothesized in our study exists. The coefficients of the quadratic and non-quadratic terms in both results confirm the presence of a non-linear relationship. The Lind Mehlum U test results and Table 4.4 confirm the presence of the Inverse U-shape relationship between Loan growth and profitability. Table 4.5 shows the U-shaped relationship between the loan growth and the NPLs.

4.2.2 Mediation Analysis

The regression results in Tables 4.4 and 4.5 show the relationship between the three variables. However, to establish the mediating role of the NPLs in the relationship between loan growth and profitability is measured using the mediation analysis (Iacobucci et al., 2007).

To check for mediation, we developed a Structural Equation Modelling (SEM) in Stata 16.1, as shown in Figure 4.1. The results are presented in Table 4.6. Column 1 shows the results for the whole sample, and Columns 2 and 3 show the subgroup results. We can see that the nature

of the relationship between the variables is also confirmed in this model, as was done using the SYS-GMM. The results of the mediation analysis are also shown in Table 4.6. The mediating effect of the NPLs on the impact of loan growth on profitability is confirmed for all three data sets. The results of mediation state whether the impact of the mediator is partial or complete using the Sobel test. The RIT is the ratio of the indirect effect to the total effect, and RID is the ratio of the indirect effect to the direct effect. The Mediation effect of the NPLs is complete in the case of public sector banks and partial in the case of private sector banks and all banks samples. The RID is 75 % for PSBs and 50% for the other two datasets, and the RIT is 3 times for PSBs, i.e. the mediation impact is 3 times the non-mediated or the direct impact of the IV on DV. The RIT is 1 time in the case of all banks and private sector banks sample.

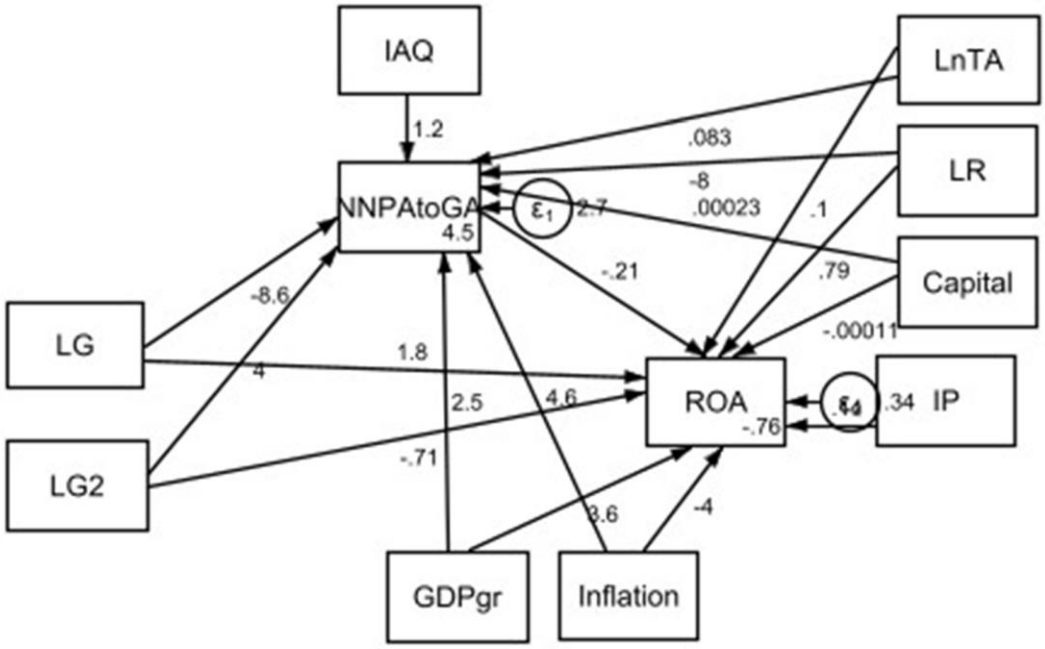

Figure 4.1 SEM Framework for Mediation Analysis.

	All Banks	Public Sector Banks	Private Sector Banks
DV- NNPA to GA			
∂GA	-8.5974***	-14.7318***	-5.7724***
∂GA^2	4.0047***	22.0363***	2.4282***
GDP	2.5222	-0.0844	0.3098
LR	-8.0134***'	0.4277***	-4.8839***
INF	4.5538	5.1754	1.4201
CAP	0.0003***	0.0001***	0.0003***
Ln TA	0.0826	-0.2528***	-0.1705***
INPA	1.1687***	2.1462***	0.7111***
_cons	4.5324***	3.1631***	5.7046***
DV- ROA			
NNPA to GA	-0.2068***	-0.1332***	-0.2757***
∂GA	1.7505***	0.6602***	1.6032***
∂GA^2	-0.7093***	0.4075	-0.6424***
GDP	3.629***	2.4495***	4.0348***
LR	0.7862	-0.2850***	0.8779
INF	-4.0275***	-4.2635***	-2.6730
CAP	-0.0001***	-0.0001***	-0.0003***
Ln TA	0.1004***	0.0854***	0.19***
IP	0.1396	0.7269***	0.1363
_cons	-0.7617***	-0.8262	-1.658
Mediation	**Partial**	**Complete**	**Partial**
RIT	**50%**	**75%**	**50%**
RID	**1 time**	**3 times**	**1 time**

Level of Sig. - ***p<0.01, **p<0.05, *p<0.1

Table 4.6 – Mediation analysis results for all bank groups

4.2.3 Quantile Regression

Table 4.7 shows the quantile regression results for all bank samples. The presence of a non-linear relationship is confirmed in all the quartiles. The Table has two parts: one shows the results for Equation 1, and the other shows the results for Equation 2. The results confirm the presence of an Inverse U shape and U shape relationship between Loan growth and profitability and Loan growth and NPL, respectively.

Table 4.8 shows the quantile regression results for the Public Sector Banks. As discussed in section 4.2.1, the results for Public Sector Banks do not confirm an Inverse U Shape relationship between Loan growth profitability; rather, they show a non-linear but U-shaped relationship. However, the quantile regression results confirm the presence of an Inverse U shape relationship in the Public sector banks in the first 3 quartiles, and then the relationship reverses. This can be explained as the large public sector banks with larger profits have a U-shaped relationship with the loan growth Whereas the relationship of the banks with small profits have an Inverse U-shaped relationship.

Similarly, Table 4.9 presents the quantile regression results for the private sector banks, confirming the SYS-GMM results. The non-linear relationship between loan growth and profitability and Loan growth and NPLs is also confirmed for the private sector banks.

	Q10	Q25	Q50	Q75	Q90		Q10	Q25	Q50	Q75	Q90
DV-ROA	**Model -13**	**Model- 14**	**Model -15**	**Model -16**	**Model -17**	**DV- NNPA to GA**	**Model -18**	**Model -19**	**Model- 20**	**Model - 21**	**Model - 22**
NNPA to GA	-0.2234***	-0.1968***	-0.2061***	-0.2337***	-0.2244***	ROA	-0.6843***	-0.9198***	-1.3105***	-1.6491***	-1.8214***
∂GA	3.1583***	1.3283***	0.9257***	0.7237***	0.5167***	∂GA	-1.5936***	-3.3238***	-3.8386***	-5.1956***	-4.8328***
∂GA^2	-3.6854***	-0.6852***	-0.3647***	-0.2102***	-0.1369***	∂GA^2	0.7171***	1.4886***	1.8765***	3.6464***	3.2701***
IP	0.4145***	0.3871***	-0.2756***	-0.0962***	-0.0837***	INPA	0.1304***	0.2440***	0.3864***	0.8231***	1.1808***
Ln TA	0.1379***	0.0868***	0.0402***	0.0075***	0.0380***	Ln TA	0.0674***	0.0050***	0.0436***	0.0838***	-0.1104***
LR	1.8926***	0.9685***	-1.0728***	0.3783***	0.2985***	LR	0.5509***	0.4235***	-1.0116***	-3.0417***	-3.8206***
CAP	-0.0002***	-0.0001***	-0.0001***	-0.0001***	-0.00016***	CAP	0.0001***	0.0001***	0.0001***	0.0001***	0.0002***
GDP	3.9450***	1.5979***	4.5132***	3.4997***	1.2539***	GDP	0.4689***	3.9882***	3.6876***	3.8214***	5.6223***
INF	-5.1417***	-5.1711***	-2.5784***	-3.5764***	-2.8855***	INF	-2.9835***	-2.7640***	-2.9707***	-1.2754***	-0.9645***

Level of Sig. - ***p<0.01, **p<0.05,*p<0.1

Table 4.7 –Regression results for effects of NPLs and Loan growth on Profitability and Loan growth on Non-Performing Loans for full sample using Quantile regression.

	Q10	Q25	Q50	Q75	Q90		Q10	Q25	Q50	Q75	
DV-ROA	**Model -23**	**Model -24**	**Model -25**	**Model -26**	**Model -27**	**DV- NNPA to GA**	**Model -28**	**Model -29**	**Model -30**	**Model -31**	**Model -32**
NNPA to GA	-0.0919***	-0.1342***	-0.1725***	-0.1883***	-0.1900***	ROA	-0.2978***	-0.6214***	-.7874***	-.7261***	-2.4102***
∂GA	1.5948***	0.6727***	-0.4920***	-1.1577***	-0.9210***	∂GA	-10.3052***	-11.3668***	-14.2287***	-14.8188***	-15.3090***
∂GA^2	-0.8530***	-0.7154***	1.3732***	3.2907***	2.9806***	∂GA^2	13.4708***	17.2274***	23.6274***	22.5144***	32.6901***
IP	1.2282***	0.6453***	0.4059***	0.3502***	0.2268***	IP	1.0611***	1.1569***	1.2022***	2.4876***	1.2459***
Ln TA	0.1660***	0.1298***	-0.0232***	-0.0154	-0.0276***	Ln TA	-0.0958***	-0.0704***	-.1136***	-.2004***	.1221***
LR	0.0407***	-0.6905***	1.2963***	-0.2557	-1.0119***	LR	2.6907***	3.1310**	3.5702***	4.1897***	-.73100***
CAP	-0.0001***	-0.0001***	-0.0000***	-0.0001***	-0.0001***	CAP	0.0001***	-0.0001***	.0001***	.0001***	-0.0001***
GDP	-1.0660***	0.3280***	3.4999***	6.2637*	1.4298***	GDP	1.7882***	0.8249	-1.0425	-13.9543***	-3.5244***
INF	-8.0492***	-1.6768***	-3.4721***	-6.9151	-2.1377***	INF	-5.2593***	-0.5738	2.4735	18.8176***	3.6166***

Level of Sig. - ***p<0.01, **p<0.05, *p<0.1

Table 4.8 –Regression results for effects of NPLs and Loan growth on Profitability and Loan growth on NPLs for a sample of Public Sector banks using Quantile regression.

	Q10	Q25	Q50	Q75	Q90		Q10	Q25	Q50	Q75	Q90
DV-ROA	**Model - 33**	**Model-34**	**Model- 35**	**Model -36**	**Model -37**	**NNPA to GA**	**Model -38**	**Model -39**	**Model- 40**	**Model- 41**	**Model- 42**
NNPA to GA	-0.6161***	-0.4301***	-0.3402***	-0.2136***	-0.3115***	ROA	-0.5367***	-0.6915***	-0.8582***	-1.2761***	-1.1272***
∂GA	2.2368***	-0.1313***	0.7411***	0.8886***	-0.3800*	∂GA	-1.7930***	-2.2948***	-1.8971***	-2.1269***	-2.3285***
∂GA^2	-3.1824***	0.1899***	-0.3215***	-0.2170***	0.3945***	∂GA^2	0.7632***	0.9377***	0.7059***	0.7893***	0.9070***
IP	-0.0486***	0.0781***	0.2129***	-0.1583***	0.8490***	IP	0.0969***	0.0995***	0.2117***	0.1982***	0.7002***
Ln TA	0.2370***	0.2412**	0.1661***	0.0509***	0.1212***	Ln TA	**-0.0001**	-0.0049***	-0.0413***	-0.0483***	-0.1915***
LR	-0.6800***	1.4047***	1.6734***	2.2040***	0.0772	LR	0.2332***	-0.5516***	-0.0589***	0.7966***	-0.6091***
CAP	-0.0003***	-0.0004***	-0.0003***	-0.0000***	0.0002*	CAP	0.0000***	0.0000***	0.0000***	0.0001***	0.0005***
GDP	1.5482***	2.8349***	0.0783	-0.6535	-2.9503	GDP	0.3301***	0.7166***	2.0621***	3.1360***	6.2277***
INF	-0.4312***	-1.4857***	-2.2260***	0.0706	-7.1448***	INF	-0.9191***	-1.4626***	-3.5957***	-5.3268***	-8.0793***

Level of Sig. - ***p<0.01, **p<0.05,*p<0.1

Table 4.9 –Regression results for effects of NPLs and Loan growth on Profitability and Loan growth on NPLs for a sample of Private Sector banks using Quantile regression.

4.3 Robust Analysis

We also check the robustness of our model by using the alternate proxies for the profitability and NPLs. ROE is a profitability ratio that uses equity instead of total assets to calculate the bank's profitability. It has been used as an alternate measure for profitability to study the relationship between loan growth and profitability and NPLs and profitability.SYS-GMM estimation model as per Models no. 2, 4, and 6 have been estimated to prove the existence of the same relationship. Further U test has been done to confirm the presence of an inverse U shape relationship.

Further, we use GNPA to GA, i.e., gross non-performing Loans to gross advances, as a measure of NPLs to test the robustness of the relationship between loan growth and NPLs. We re-estimate Model 8, Model 10, and Model 12 using the alternate variable GNPA to GA and get similar results. The results of the U test also prove a U-shaped relationship between loan growth and non-performing loans.

The results of the robust analysis are presented in Table 4.10. The results resonate with the results of our empirical analysis and thus strengthen our model. It ensures the validity of our results.

Notes to the Table - The Values in Bold are insignificant

Table 4.10 Robust Analysis

	All Banks Sample	PSB	Pvt.Bank		All Banks	PSB	Pvt.Bank
DV-ROE	**Coef.**	**Coef.**	**Coef.**	**DV- GNPAtoGA**	**Coef.**	**Coef.**	**Coef.**
NNPAtoGA	-3.7620	-2.4041	-3.7743	**ROA**	-0.0192	-0.0146	-0.0201
∂GA	11.8705	21.0156	15.9505	∂GA	-0.0716	-0.2796	-0.0573
∂GA^2	-6.9000	**-31.1166**	-7.9106	∂GA^2	0.0257	0.5973	0.0218
IP	0.4651	1.3219	0.2521	IP	0.6715	0.7940	0.2815
LnTA	2.1684	1.0862	1.5439	LnTA	**0.0025**	**0.0063**	-0.0021
LR	**0.3973**	**-8.0841**	24.4820	LR	-0.1886	**-0.1497**	-0.0472
LnCAP	-0.0018	-0.0019	-0.0021	LnCAP	0.0000	0.0000	0.0000
GDP	26.8177	**7.5016**	32.3823	GDP	**0.0069**	**-0.0026**	0.0480
INF	-36.8629	-97.8543	**-25.0807**	INF	**0.0086**	**-0.0170**	**-0.0094**
_cons	-11.2472	**2.1732**	-19.1579	_cons	0.1192	**0.0575**	0.0903
Hansen Sargan Test	0.2790	0.7870	0.7760	Hansen Sargan Test	0.1900	0.5430	0.6700
diff	0.0210	0.0650	0.0740	Arellano Bond test 1st diff	0.1020	0.0780	0.1220
diff	0.4550	0.3030	0.0830	Arellano bond test 2nd diff	0.1670	0.5620	0.3280
No.of Observations	464	192	272	No.of Observations	464	192	272
U shape test	Inverse U shape	U shape	Inverse U shape	U shape test	U shape	U Shape	U Shape
Extreme point	0.8602	0.3111	1.0082	Extreme point	1.3933	0.2341	1.3123
Lower Bound	-0.2901	-0.2256	-0.2901	Lower Bound	-0.2901	-0.2256	-0.2901
Upper bound	2.1632	0.5628	2.1632	Upper bound	2.1632	0.5628	2.1632

4.5 Summary & Hypothesis Validation

Table 4.11 shows the summary of the results of our analysis. The Table summarises the results presented in Tables 4.4 and 4.5.

	All Banks	PSB	Pvt Banks		All Banks	PSB	Pvt Banks
DV-ROA				**DV-NNPAtoGA**			
NNPAtoGA	-ve	-ve	-ve	**ROA**	-ve	-ve	-ve
∂GA	+ve	-ve	+ve	**∂GA**	-ve	-ve	-ve
∂GA^2	-ve	+ve	-ve	**∂GA^2**	+ve	+ve	+ve
IP	+ve	N.S	+ve	**INPA**	+ve	+ve	+ve
LnTA	+ve	+ve	+ve	**LnTA**	+ve	N.S	+ve
LR	N.S	N.S	-ve	**LR**	-ve	N.S	-ve
LnCAP	+ve	+ve	+ve	**LnCAP**	NS	+ve	-ve
GDP	+ve	+ve	+ve	**GDP**	+ve	N.S	+ve
INF	-ve	N.S	-ve	**INF**	-ve	N.S	N.S

Table 4.11 Summary of the results

After discussing the results of our analysis, we move to hypothesis validation to finally validate the theoretical framework we developed in Chapter 3. We use SYS-GMM, Lind Mehlum U test, and Mediation analysis using SEM to prove our hypotheses. All four hypotheses of our study are accepted and supported by the results of the empirical analysis. Table 4.12 is the hypothesis validation table that shows a summary of the hypotheses of our study and its validation as per the results in Tables 4.4,4.5 and 4.6

Variables	Nature of relationship	Hypotheses
LG → NPL	U shape (SYS-GMM &Utest)	H1 - accepted
LG → Profit	Inverse U shape (SYS-GMM &Utest)	H2- accepted
NPL → Profit	Negative (SYS-GMM)	H3- accepted
LG → NPL → Profit	NPL mediates the relationship between LG and Profit (SEM and Mediation Analysis)	H4- accepted

Table 4.12 Hypothesis Validation

Chapter 5
Conclusion

Chapter 5

Conclusion

The study results show that steady loan growth may be profitable for the banks as an increase in loans increases profitability and reduces NPLs. However, an accelerated level of loan growth starts to negatively impact profitability as the banks' asset quality starts deteriorating. Many theories can explain the reason behind these variables' behaviour in existing literature regarding management capability. Overburdened employees due to excessive loan growth may lead to bad decision-making and riskier asset accumulation. (Kashif et al., 2016) their study concluded that excessive loan growth and weak regulations increase bank risk. Hence, in times of excessive loan growth, the regulators must be vigilant as to whether all the norms are being followed or not to ensure the quality of loans does not deteriorate.

The result of our study explains that excessive lending deteriorates asset quality, which in turn affects profitability, thus defeating the whole purpose of increased lending. Positive manifestations of loan growth in the Indian banking sector include increased investment, higher consumer spending, job creation, and overall economic growth. This, in turn, can lead to increased profitability for banks. However, excessive loan growth can also result in negative manifestations, such as a higher incidence of non-performing loans, adversely affecting bank profitability and stability.

Therefore, striking a balance between loan growth and non-performing loans is crucial for banks to maintain sustainable profitability while ensuring the financial system's stability. This balance can be achieved by implementing effective risk management practices, such as conducting thorough credit assessments and establishing adequate loan loss provisions.

Additionally, encouraging responsible borrowing by promoting financial literacy and ensuring borrowers can repay their loans can help minimize non-performing loans and promote long-term profitability in the banking sector.

Loan growth and Profitability reveal an Inverse U shape relationship in our study. Incorporating the quadratic term for the loan growth mimics excessive loan growth. In the initial phase of loan growth, the profitability curve is increasing, but if the lending speed is too fast, the profitability curve starts to change.

The change of the course profitability curve about the loan growth can be understood in the following stages-

1- The phase of growth

Loan growth is linked to an increase in interest income. The loans are treated as an asset in a bank's balance sheet. The interest they earn is the major source of bank income. Thus, loan growth is also primarily profitability growth.

2- The phase of speedy lending

Lending money and accepting deposits is a bank's primary activity. But unlike accepting deposits, lending money to borrowers is not simple. Lending requires a lot of pre-sanction and post-sanction follow-up. Sanctioning a loan and monitoring its performance until it is finally repaid requires specialized skills and time devoted by the loan officials. Banks sometimes decide to speed up the pace of lending for some reason. The increased speed of lending can have many reasons - due to some special targets given to the bank, the bank's decision to strengthen the asset side of the balance sheet. Whenever there is pressure to increase the lending speed, it manifests as deteriorated asset quality. A speedy loaning process leads to lapses in the pre-sanction process, which involves assessing the credit needs of the borrowers,

their creditworthiness, and the collateral valuation. Any lapse in this part creates a risk of future loan delinquency.

3- The phase of decline in profitability

After extending the loan, a loan may turn bad immediately or at any stage. The bank officials have the task at hand to monitor the health of the loan till the time it is fully repaid. Loans have an inherent risk of default even when all the presanction appraisal is done as per the guidelines. The fast pace of lending also increases the NPLs, as proven in our study. Various reasons for this can be stated as under:

Credit Risk – As the speed of lending increases, the risk for default by the borrower may increase, which also increases the provision required for doubtful debts and NPAs thus reducing profits.

Operational Challenges- Managing a growing loan portfolio can become a challenge for the bank, which also affects the bank's profitability.

Demand -Supply mismatch- There is a limited demand for bank finance in every market. Beyond this point, the money advanced by the bank may be diverted and not be put to proper use, thus leading to non-repayment of loan.

Mediating role of NPLs

- Non-performing loans are an intermediary, a critical risk element connecting loan growth to profitability. If a bank's loan portfolio grows too quickly without proper risk management, the growth in non-performing loans can erode profitability.

- The mediation process occurs as follows: aggressive loan growth may increase non-performing loans, triggering increased provisions for loan losses. Increased provisioning directly influences the bank's profitability by lowering net income.

Whenever banks try to increase loan growth, banks must focus on mitigating the risk also:

1. Emphasis should be on maintaining a balanced and long-term strategy for loan portfolio management.
2. Banks can use risk management measures to break or mitigate the mediation effect. This may include tightening underwriting requirements, upgrading risk assessment systems, and actively maintaining the loan portfolio's credit quality.

In summary, non-performing loans are important in moderating the relationship between loan growth and profitability. The challenge for banks is to strike the right balance between loan growth and profitability without jeopardizing asset quality. Effective risk management strategies and early actions to address non-performing loans are critical to a banking institution's long-term financial health and viability.

Chapter 6
Implications, Limitations and Future Research Direction

Chapter 6

Implications, Limitations and Future Research Direction

6.1 Implications of the Study

As already discussed in the introduction section, Indian banks have a very important role in the nation's economic growth. However, the banks must maintain their health by balancing growth and safety by adhering to steady loan growth. (Duho et al., 2020) also suggests that having a proper mix of assets in the loan portfolio of the banks also has a positive impact on the performance of the banks in emerging economies. Our study results highlight this trade-off by suggesting a theoretical framework and later proving it empirically. The study has many theoretical and practical implications, which this chapter discusses.

6.1.1 Theoretical Implications

Our study links the four hypotheses put forward by (Berger et al., 1997) with loan growth. The framework developed links these hypotheses with loan growth. The result of our empirical analysis reveals that the fast pace of lending increases the risk of NPLs, which can be explained by the “Bad Management”, “Skimping,” and “Moral Hazard” hypotheses. With the increased pressure of work, the loan officials and management may compromise on the appraisal part of the loans due to lack of time. They may also lax the standards for lending to increase the quantum of loans. This increases the problem of bad loans. The theory given by (Berger et al., 1997) links the problem loans with the cost efficiency using these hypotheses. We link these hypotheses with the speed of lending and use these to empirically test the relationship between loan growth, NPLs, and profitability. The study results can provide a base for these hypotheses regarding management behaviour. Our study proposes that

managers are not inherently incompetent or have skimping behaviour; it is induced by the lack of time or increased work pressure due to the fast pace of lending. This bad decision or hasty work by the loan officials translates into bad loans and subsequently lower profitability. We also link the non-linear relationship between our variables with the business cycle. The Business cycle has four phases: expansion, boom, recession, and depression. During the phase of expansion and boom, the loan growth increases due to an overall boost in economic activities. The boom phase is followed by a recession after a point. During the recessionary pressure, the loans need a lot of monitoring as the borrowers face financial stress and struggle to survive. The value of collaterals also starts to diminish in these times. Thus, banks must keep a close vigil on the repayment of loans. Recession and depression phases are accompanied by a phase of increasing NPLs and decreasing profitability. The increased lending in the periods of expansion and boom should not compromise the credit appraisal and monitoring aspects. Any laxity in this part may bring in more profits for the banks but will translate into problem loans in times of recession. Thus, we explain the hypotheses of the earlier study by linking it to the speed of lending and the business cycle.

6.1.2 Managerial and Policy Implications

The findings of this study have immense importance for the managers of banks and the policymakers. Senior management and regulatory authorities always pay a lot of attention to compliance with all the rules and regulations to ensure no lapses in sanctioning loans through inspections and audits. However, there is a need to –

- Assess the time and manpower required to process a loan. The targets for loan disbursement should also be fixed considering this aspect. Disregarding this aspect of

lending while allotting budgets and targets to the branches creates room for relaxing the standards or lapses in the pre-sanction and post-sanction parts of the loaning process due to excess work pressure. These lapses translate into bad loans.

- The performance-linked incentives to the employees should be linked to the quality of the loans sanctioned rather than the number of loans sanctioned.
- Authorities should always monitor the speed of lending by the branches.
- Lending is a routine activity for banks, but requires specialized knowledge, thus, banks must ensure that bank officials dealing with loans should be qualified to deal with it.
- Senior management and the loan official must be vigilant about the economic conditions in their service area to understand and thus decide the future course of action for the branch.

6.1.3 Social Implications

The impact of fast-paced lending and rising NPLs is far-reaching for a developing nation. Bank finance is a very important source of finance for all economic activities. A small shop owner to a large-scale industry all need bank finance for their business needs. When banks choose speedy lending, there are a lot of instances when a bank overfinances a project. Overfinancing is very harmful to both the borrower and the bank. When the borrower has more money than the actual requirement of the business, there is a huge risk of money being diverted to the other needs of the borrower. Such diversion of funds makes it difficult for the borrower to repay of loan. A default in repaying a bank's instalment makes the borrower a defaulter and negatively impacts the borrower's credit rating for future loan requirements. This creates a very negative environment for bank lending. Increasing NPLs is harmful to both the banks and the borrowers. Mounting pressure on bank profitability due to rising NPLs

discourages the banks, also it creates a bad bank rating, which is also harmful for the shareholders and investors.

6.2 Limitations and Future Research Direction

The study is confined to only the Indian banking sector, it can be extended to other similar Asian economies where banks play an important role in implementing governments' growth objectives through priority sector lending like Indonesia, Malaysia, the Philippines, Thailand, and Vietnam.

The model can also be empirically tested in any other economy as the cyclical nature of the business cycle exists globally, which is the driving force behind this model.

Also, other profitability and non-performing loan indicators can be included in the study to improve the robustness study such as net interest margin, Z-Score for profitability, and loan loss provision as an indicator of non-performing loans.

This study can also be strengthened by adding a qualitative analysis of the impact of increased workload on employees due to fast-paced lending and its impact on their performance.

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Appendix

List of Publications

1. Gupta, S., Bansal, R., Kar, S., and Mishra, S., (2019), "Portfolio Management for Amateur Investors, A case study", *The Case Centre, Cranfield University, Wharley End Beds MK43 0JR, United Kingdom* (UK)

2. Bansal, R., Kar, S., & Gupta, S. (2021). Efficiency assessment of consumer's electronics sector: Data envelopment analysis. *Journal of Asia-Pacific Business, 22*(4), 279-297.DOI 10.1080/10599231.2021.1983502 ABDC C I.F 1.4

3. Gupta, S., & Bansal, R. (2024). Manifestations of loan growth–profitability and non-performing loans–a study of the Indian banking sector. *South Asian Journal of Business Studies.* DOI 10.1108/SAJBS-04-2023-0100 ABDC C I.F 2.4

4. Gupta, S., & Bansal, R. (2024). Understanding bank lending and its relationship with profitability and non-performing loans: A meta-analysis. *Journal of Economic and Administrative Sciences.* (Accepted) ABDC C I.F 1.8

5. Bansal, R., Verma, R., & Gupta, S. (2024). The Energy Market: Transformation, Privatization, and Diversification, The South Asian Energy Corridor Toward Sustainability and Security. Apple Academic Press **ISBN:** 9781774914786 (In Press)

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Shweta Gupta is a research scholar at Rajiv Gandhi Institute of Petroleum Technology, Uttar Pradesh. Her research interests are Banking, Credit risk management and NPA management.

Her thesis explored various components related to studying the impact of loan growth on profitability with the mediating role of NPAs to provide a framework that can be used as a reference for framing policies for credit expansion. She has published 3 research articles in high-indexed SCOPUS and ABDC-ranked journals. She has also published one case study and one book chapter. She also reviews research papers for prominent journals like the South Asian Journal of Business Studies and Economic and Administrative Sciences. She has completed her M.com at the University of Delhi. She has work experience of 7 years in Punjab National Bank as a Sr. Manager.